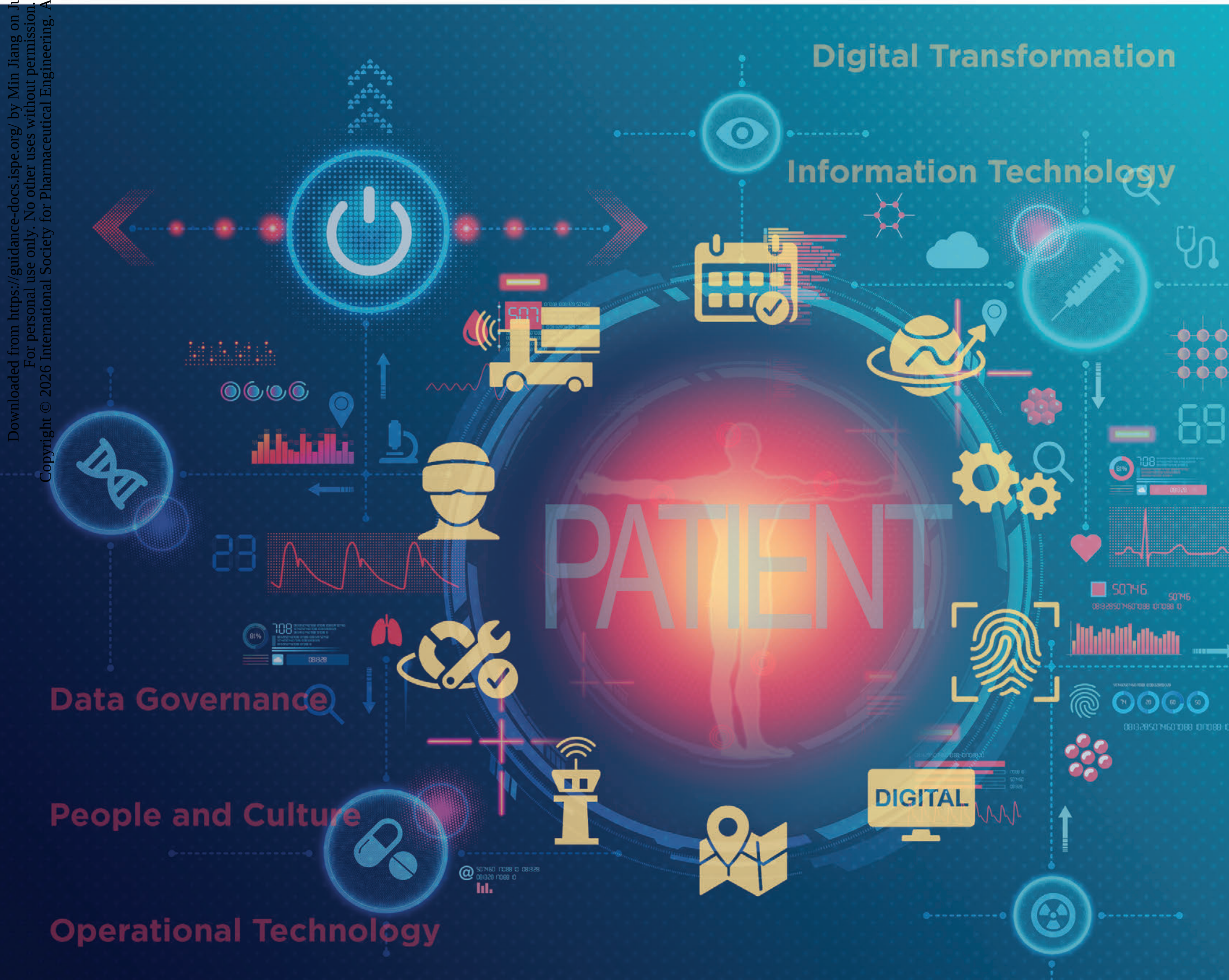
ISPE | PHARMA 4.0™

GOOD PRACTICE GUIDE:

Pharma 4.0™ – Holistic Digital Enablement



ISPE® | PHARMA 4.0™

GOOD PRACTICE GUIDE:

Pharma 4.0™ – Holistic Digital Enablement

Disclaimer:
This Guide provides practical guidance on holistic digital enablement to support the implementation of Pharma 4.0™. This Guide is created and solely owned by ISPE. It is not a regulation, standard or regulatory guideline document. ISPE cannot ensure and does not warrant that a system managed in accordance with this Guide will be acceptable to regulatory authorities. Further, this Guide does not replace the need for hiring professional engineers, technicians, or consultants.

Limitation of Liability
In no event shall ISPE or any of its affiliates, or the officers, directors, employees, members, or agents of each of them, or the authors, be liable for any damages of any kind, including without limitation any special, incidental, indirect, or consequential damages, whether or not advised of the possibility of such damages, and on any theory of liability whatsoever, arising out of or in connection with the use of this information.

ISBN 978-1-946964-99-1

Preface

The pharmaceutical industry stands at a pivotal moment—facing the immense promise of digital transformation in alignment with Pharma 4.0, while navigating the complex challenges that come with it. As digital technologies rapidly evolve, the need for a coordinated, strategic, and human-centric approach to digital enablement has never been greater.

This *ISPE Good Practice Guide: Pharma 4.0™ – Holistic Digital Enablement* has been developed by the Holistic Digital Enablement Subcommittee of the ISPE Pharma 4.0 Community of Practice (CoP) to support organizations across the pharmaceutical value chain. It is designed for a broad audience—spanning operations, quality, compliance, regulatory affairs, Information Technology (IT)/Operational Technology (OT), manufacturing, Research and Development (R&D), supply chain, Human Resources (HR), marketing, sales, and data governance. Regardless of your role within the organization or which phases you are involved in, if you have a stake in digital transformation, this Guide is for you.

Pharma 4.0 is more than a technological shift; it is a holistic transformation of culture, capabilities, and connectivity. It promises to enhance efficiency, reduce costs, increase quality, and ultimately, improve patient outcomes. But realizing this vision requires more than deploying new tools—it demands strong leadership, cross-functional collaboration, and robust frameworks for data governance, cybersecurity, and change management.

Emerging technologies—such as Artificial Intelligence (AI), Machine Learning (ML), the Internet of Things (IoT), Process Analytical Technology (PAT), blockchain, and big data analytics—are reshaping the pharmaceutical landscape. These innovations enable smarter, faster, and more flexible operations, from predictive maintenance to real-time quality monitoring. However, without the right foundation—people, processes, and governance—these tools alone cannot deliver sustainable transformation.

This Guide focuses on critical enablers for holistic digital success, including knowledge management, critical thinking, data governance, standardization, and the human element. It emphasizes that digital transformation is not solely a technical challenge, but also a cultural one. Organizations must cultivate a mindset of continual learning, empower their workforce, and create an environment where innovation can thrive.

More importantly, this Guide is not a rigid, step-by-step playbook. Instead, it offers flexible, best-practice-driven strategies to address a variety of problem statements identified by the authoring team, that can be adapted to an organization's unique culture, technological landscape, and digital maturity. It aligns with other ISPE Guidance Documents, including the *ISPE GAMP® 5: A Risk-Based Approach to Compliant GxP Computerized Systems (Second Edition), ISPE GAMP® Good Practice Guide: IT Infrastructure Control and Compliance (Second Edition),* and *ISPE APQ Guide Series: Advancing Pharmaceutical Quality – Cultural Excellence,* offering readers a comprehensive and interconnected framework.

As the industry continues to evolve, this Guide serves as a practical resource to help stakeholders not only keep pace with change, but to lead it. Through thoughtful planning, cross-disciplinary collaboration, and a commitment to both innovation and compliance, organizations can overcome key barriers—from cybersecurity risks and regulatory hurdles to workforce adaptation and legacy infrastructure.

By embracing a holistic approach to digital enablement, we can collectively shape a smarter, more resilient pharmaceutical industry that delivers greater value to patients and society. We hope this guide provides the insight, inspiration, and practical direction you need to successfully advance your digital transformation journey.

Acknowledgements

The Guide was produced by an international Task Team, under the leadership and direction of:

Nuha Al-Hafez	FUJIFILM Biotechnologies	USA
Yvonne Duckworth, PE	CRB	USA
Jason Schneider	AbbVie Inc.	USA

The work was supported by the ISPE Pharma 4.0™ Community of Practice (CoP).

Team

The following individuals assisted in the preparation of this Guide:

Bryon Hayes, P.Eng.	Grantek	Canada
Ryan Thompson	CRB	USA
Stephen Wing	MilliporeSigma	USA
Dirk Wollaert	Siemens	Belgium
Vinzenz Zauner	Zauner Consulting	Germany

Special Thanks

We extend a special thank you to Thomas Zimmer (ISPE Vice President, European Operations, Germany) for his advice during preparation of the Guide.

The Team thanks ISPE for technical writing, editing, and project management support by Gabriela Kantor and Nina Wang (ISPE Guidance Documents Technical Editors) and production support by Lynda Goldbach (ISPE Senior Publications Manager).

The Team would like to express their grateful thanks to the many individuals and companies from around the world who reviewed and provided comments during the preparation of this Guide; although they are too numerous to list here, their input is greatly appreciated.

Company affiliations are as of the final draft of the Guide.

Cover photo: www.istockphoto.com.

701 N. Westshore Blvd., Suite 207, Tampa, Florida 33609 USA
Tel: +1-813-960-2105, Fax: +1-813-264-2816

www.ISPE.org

Table of Contents

1 Introduction

1.1 Background

Pharma 4.0 presents transformative opportunities to enhance efficiency, quality, cost-effectiveness, and ultimately, patient outcomes. However, seizing these opportunities comes with its own set of challenges, including regulatory compliance, cybersecurity risks, legacy infrastructure, workforce readiness, and data governance. With strategic planning, digital transformation can drive innovation and operational excellence across the industry. For an introduction on how the concept of Pharma 4.0 was developed from Industry 4.0, refer to the *ISPE Baseline® Guide: Pharma 4.0*™ [1].

Pharma 4.0 builds upon emerging technologies such as Artificial Intelligence (AI), Machine Learning (ML), the Internet of Things (IoT), Process Analytical Technology (PAT), blockchain, and big data analytics, which are rapidly reshaping how pharmaceutical organizations operate. A few concrete examples of using these technologies include the use of AI for predictive quality control, blockchain for supply chain traceability and for ensuring tamperproof audit trails for drug provenance, and PAT for enabling real-time quality monitoring during manufacturing. These technologies enable real-time monitoring, predictive maintenance, and advanced analytics that drive more informed, data-driven decision-making and more agile operations. Digital technologies are covered in *ISPE GAMP® 5: A Risk-Based Approach to Compliant GxP Computerized Systems (Second Edition)* [2], for example, Appendix D10 – Distributed Ledger Systems (Blockchain) and Appendix D11 – Artificial Intelligence and Machine Learning (AI/ML).

1.2 Purpose

The purpose of this Guide is to provide a comprehensive framework and a set of good practices for the industry, helping lead and shape the future of pharmaceutical manufacturing in the Pharma 4.0 era. It aims to accelerate digital implementation by helping translate business and compliance requirements into the following:

- Actionable and practical strategies, suitable for organizations at different levels of digital maturity for Information Technology (IT), Operational Technology (OT), and digital platforms.
- Detailed clarification of benefits and challenges related to emerging technologies, offering insights to better understand their application and impact

.

- An enhanced data governance framework to ensure compliance with current and future regulatory standards, addressing contributions amidst emerging technologies.

Together, these elements can guide organizations in effectively navigating the complex intersection of digital innovation, compliance, and data integrity.

1.3 Scope

This Guide focuses on holistic digital enablement for the pharmaceutical industry, including specific use cases. The topics of GAMP, GMP, Quality Risk Management (QRM), and validation are outside the scope of this Guide and are comprehensively addressed in other relevant guides and regulations.

Rather than serving as a prescriptive, step-by-step manual, this Guide provides agnostic insights and strategies by offering flexible, best-practice-driven approaches. These approaches are adaptable and require tailoring to an organization's unique culture, digital maturity, and technological environment.

Furthermore, this Guide exists in context with other documents. Where possible, references to other in-depth guides are given. Key references that provide a practical set of guidelines and principles to ensure the quality and integrity of computerized systems used in regulated industry are *ISPE GAMP 5 Guide (Second Edition)* [2] and *ISPE GAMP Good Practice Guide: for IT Infrastructure Control and Compliance (Second Edition)* [3]. Further insights on culture challenges faced by industry are discussed in the *ISPE Guide Series: Advancing Pharmaceutical Quality – Cultural Excellence* [4].

1.4 Key Concepts

This Guide focuses on critical elements such as Knowledge Management (KM), critical thinking, data governance, and the human element including people and culture in Pharma 4.0. Efficient KM and data governance practices are essential for maintaining regulatory compliance and driving innovation. As data-driven decision-making becomes more prevalent, it is critical to have robust network infrastructure to support IT/OT systems by providing seamless integration across all systems and production stages. Equally important is fostering a culture of continual learning and innovation to empower the workforce in embracing new technologies.

This comprehensive approach will equip stakeholders with the insights needed to navigate the evolving landscape and capitalize on the potential of emerging technologies.

1.5 Key Terms

ALCOA++: ALCOA++ is a set of data integrity principles, built upon the original ALCOA principles, that guide how data is collected, managed, and stored to ensure its reliability and trustworthiness. These principles are crucial for compliance with GMP and other regulatory requirements. ALCOA++ stands for Attributable, Legible, Contemporaneous, Original, Accurate, Complete, Consistent, Enduring, Available, Traceable.

Business Process Management (BPM): A systematic approach to improving an organization’s processes, making them more efficient and effective through continuous optimization and automation.

Data: Raw, unprocessed facts and figures collected from various sources, which can be in the form of numbers, text, images, or other types of information. Data serves as the foundation for analysis, decision-making, and generating insights in various fields.

Digital Twin of the Organization (DTO): A virtual model of an organization’s processes, systems, and assets, used to simulate, analyze, and optimize performance in real-time.

Enterprise Architecture Management (EAM): The practice of analyzing, designing, planning, and implementing enterprise analysis to successfully execute on business strategies, ensuring alignment between IT and business goals.

2 Problem Statement

Realizing the benefit and technologies of Pharma 4.0 requires overcoming challenges such as dynamic technology, an evolving regulatory landscape, a lack of data governance, workforce skill gaps in Pharma 4.0, and challenges with legacy systems, as well as with data management, data standards and formats, and cybersecurity. These barriers hinder consistent control and collaboration and are explored in detail throughout this chapter.

In addition, it is important to consider the creation of a business continuity plan for navigating the complexities and challenges of digital enablement and ensuring long-term success.

2.1 Dynamic Technology and Regulatory Landscape

The pharmaceutical industry is evolving rapidly in the Pharma 4.0 era, driven by the integration of emerging technologies and digital transformation. This shift from traditional manufacturing, quality, and regulatory practices to a more interconnected and data-driven approach aims to enhance efficiency, quality, and compliance. Despite technological advancements, industry continues to face challenges, particularly a lack of effectively managing and regulating these new technologies.

The existing regulatory framework, designed for conventional practices, must adapt to accommodate the aspects introduced by digital transformation. This adaptation is ongoing, with both regulators and pharmaceutical organizations learning and evolving their approaches in real time to navigate these changes. Initiatives such as the US FDA's Emerging Technology Program [5] and EMA's Innovation Task Force [6] and Quality Innovation Group [7] are facilitating dialogue between regulators and industry to provide early regulatory support, clear guidance, and streamlined pathways. These efforts are intended to assist pharmaceutical organizations in adopting innovative manufacturing technologies, improving product quality, and accelerating time to market.

In the era of Pharma 4.0, proactive quality management is essential to ensure product integrity and regulatory compliance in increasingly complex manufacturing environments. Digital tools—including visualization techniques such as scatter plots and Statistical Process Control (SPC) charts—play a critical role in tracing Out of Specification (OOS) events back to their root causes. As data volumes grow, these tools can help uncover hidden patterns and signals that might otherwise be obscured. To fully realize the benefits of Pharma 4.0, it is vital that digital systems support robust root cause analysis and continual improvement. Aligning these efforts with ICH Q10 [8] principles ensures a lifecycle approach to quality management that fosters innovation and operational excellence.

2.2 Handling Increasing Amounts of Complex Information

One aim of Pharma 4.0 is to embed knowledge into automated systems, for production processes and business processes alike. As digitalization efforts intensify, the level of detail required for process knowledge increases substantially. Full process understanding includes detailed steps, deviations, trigger points, and alternative pathways. If available at all, this knowledge is fragmented across the organization. Holistically managing the vast amount and complexity of information emerges as a key challenge.

Traditional documentation methods for manual operations—such as Standard Operating Procedures (SOPs), wiki-based knowledge bases, and intranets—are designed for human readers and users. These formats rely on expert understanding and the unparalleled contextualization skills of the human mind. However, they lack the comprehensive and structured information needed for future digitalization projects.

2.3 Lack of Data Governance

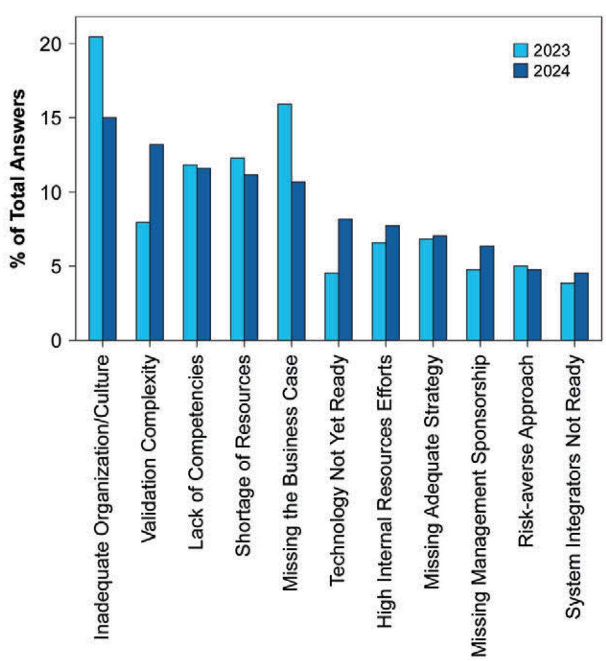
The lack of standardized data governance in the pharmaceutical industry impacts decision-making, regulatory compliance, and product integrity. Fragmented systems, inconsistent data entry, and decentralized repositories lead to poor data quality, regulatory risks, and potential breaches. Noncompliance with Pharmaceutical Inspection Co-operation Scheme (PIC/S) [9] standards such as PI 041-1 (Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments) [10] can result in product recalls, fines, and loss of market access. To ensure patient safety and trust, organizations must implement robust governance frameworks, enforce data integrity measures, standardize SOPs, and provide compliance training.

Regulations and guidelines such as 21 CFR Part 11 [11], EU EudraLex Volume 4 Annex 11 and Annex 15 [12, 13], and ICH Q10 [8] require data integrity to uphold product quality and safety. Emerging technologies such as AI and ML enhance drug development but also introduce risks, including biased data and algorithmic opacity. Additionally, the *ISPE GAMP® Guide: Records and Data Integrity* [14] provides practical guidance to support data integrity within regulated environments. Ensuring compliance requires rigorous validation, strong governance frameworks, and comprehensive risk assessments per ICH Q9(R1) [15]. Transparent, accountable, and continuously monitored AI-driven systems can allow organizations to harness innovation while maintaining compliance and data integrity. Adopting the principles of data integrity by design during system development ensures ALCOA++ compliance from the outset, reducing retrospective remediation costs.

2.4 Workforce Skill Gaps in the Pharma 4.0 Era

The pharmaceutical industry is at a pivotal crossroads and facing significant workforce skill gaps as it transitions into the Pharma 4.0 era. The integration of emerging digital technologies such as AI, ML, and advanced robotics presents both opportunities and challenges. Although these technologies promise to revolutionize operations, there is a growing disparity in the skill sets required. The industry needs professionals with expertise in emerging technologies, including data scientists, AI specialists, and cybersecurity experts, but there is a shortage of these talents. These new professionals with emerging technology expertise may be coming from another industry, so they will require pharmaceutical training. According to the ISPE Pharma 4.0 Annual Survey [16], the challenge of "lack of competencies" has been consistently high and stagnant since 2021 and still continues to be the case in 2023 and 2024. Refer to Figure 2.1 for results from 2023 and 2024.

Figure 2.1: Perceived Challenges to Pharma 4.0 – Results from 2023 and 2024



Additionally, current employees deeply rooted in traditional practices must undergo upskilling to adapt to new requirements. Organizations must invest in comprehensive training programs focused on digital manufacturing, AI/ML, data analytics, and cybersecurity to ensure their workforce is equipped for digital transformation. In addition, leveraging resources such as specific AI and data science courses, vendor-specific certifications, and ISPE's training courses and webinars can help to accelerate workforce upskilling.

2.5 Knowledge Management and Training

Another critical issue is the preservation of tacit knowledge, which is at risk of being lost during this transition. Tacit knowledge refers to the knowledge gained from personal experience. It is often referred to as "know-how" and it is understood without being formally documented. This knowledge is an asset that cannot be easily transferred or replaced by new technologies. To sustain this tacit knowledge or know-how, organizations must foster a culture of continual learning and mentorship. Cross-functional teams that blend experienced professionals with tech-savvy employees can facilitate the transfer of tacit knowledge while integrating new digital skills. This approach helps bridge the gap between current capabilities and future needs, ensuring that organizations can maintain their competitive edge while embracing digital transformation.

Furthermore, upskilling should focus not only on individuals but also on team depth and breadth, with development opportunities and assignments that drive the organization forward. Cross-training is also key, as the success of digital transformation can be improved when organizations divest cross-training within these cross-functional groups. This can result in experienced professionals becoming more tech-savvy, and the tech-savvy employees can gain tacit knowledge of the industry. A concrete example of specific cross-training is between Quality Assurance (QA) and data scientists. This cross-training can help to foster a shared language, improve collaboration on data-driven quality initiatives, and enable both groups to jointly develop and validate advanced analytics and digital tools. Another example of cross-training involves data scientists/engineers understanding laboratory workflows and the inherent challenges, while analytical scientists needing to be willing and able to see the benefits of digitalization (and to break the cycle of paper-based processes).

To support these initiatives, KM must be prioritized as a foundational pillar. Effective KM systems enable the collection and dissemination of both explicit and tacit knowledge (or institutional knowledge). This supports real-time decision-making, business process understanding, continual improvement, and regulatory compliance. In the context of Pharma 4.0, KM should be dynamic, integrated into the organization's operations, and capable of capturing undocumented processes. By advancing KM, organizations can mitigate the risks of knowledge loss due to employee turnover and foster a culture of innovation. A robust KM strategy also streamlines regulatory processes, facilitating quicker approvals and enhancing operational efficiency. Addressing skill gaps, preserving tacit knowledge, and implementing strong KM systems will enable the pharmaceutical industry to navigate the transformation successfully and ensure adaptability in a rapidly evolving landscape. To learn more about tacit knowledge and KM tools and processes, refer to the *ISPE Good Practice Guide: Knowledge Management in the Pharmaceutical Industry* [17].

Additionally, the pharmaceutical industry faces silos that hinder collaboration, efficiency, and innovation. Regulatory agencies have differing requirements, creating complex compliance challenges that delay global product approvals. Internally, departments such as R&D, quality, operations, and regulatory affairs often operate independently, leading to inefficiencies and misaligned priorities. Collaboration between not only industry and regulatory agencies, but also within organizations, can present many challenges. It is important to consider how collaboration can be enabled, given other challenges mentioned in this chapter.

There is also a lack of clarity around how digital innovations will be received by regulatory agencies and the mechanisms for introducing new technologies. Moreover, silos extend beyond industry, with academia, industry, and regulatory bodies working in isolation in some cases, limiting the exchange of knowledge and slowing progress in research, regulatory science, and patient access to treatments. Addressing these silos requires harmonizing regulatory standards, fostering cross-sector collaborations, and promoting shared goals. In addition, the benefits of breaking down organizational silos include the improvement of overall communication, agility, and adaptability.

2.6 Challenges with Legacy Environments

In pharmaceutical manufacturing, legacy systems typically refer to older hardware and software platforms (often decades old) that remain in use despite being technologically outdated. These systems are often retained because replacing them can be costly and disruptive to critical operations and processes. It is often more practical to maximize the return on the original investment made for the older technologies. Common examples include early versions of Distributed Control Systems (DCS), stand-alone Programmable Logic Controllers (PLCs), and custom-developed Supervisory Control and Data Acquisition/Manufacturing Execution System (SCADA/MES). Local data loggers or chart recorders are still widely used, as are stand-alone laboratory instruments such as High-Performance Liquid Chromatography (HPLC), Gas Chromatography (GC), and balances. Many of these systems were implemented before data integrity requirements (according to 21 CFR Part 11 [11] and ALCOA++) and therefore they may not meet current compliance expectations. Specific examples include the Siemens S5 PLCs, which are obsolete and difficult to support, and DOS-based Human Machine Interfaces (HMIs) that are incompatible with modern operating systems. A particularly widespread example of legacy processes is the continued use of paper-based batch records, which lack digitization and are highly prone to human error and data loss.

Legacy systems often lack the appropriate technical controls to ensure data integrity. Raw data, if stored at all, is typically stored in a proprietary format. If it can be exported, it is likely to be in text or Comma-Separated Value (CSV) format. Those file formats are easy to manipulate, and do not comply with ALCOA++ principles. These systems were not designed to be integrated in a connected environment. As such they are often not integrated at all or the integrations created to access data from these systems are complex, fragile, and insecure. To support these systems, manufacturers often rely on “heroes,” Subject Matter Experts (SMEs) with system knowledge that is tacit, rather than comprehensive and robust processes that spread the knowledge throughout the organization.

Modernization strategies, such as wrapping legacy systems with Application Programming Interfaces (APIs) or adopting microservices, can enhance interoperability and data accessibility while maintaining compliance with ALCOA++ principles.

Additional primary challenges with legacy systems, focused on business continuity and transitions and migrations, include the following:

- Business disruption caused during transitions
- Possibility of data loss during transitions
- Challenges with data-intense systems during migrations
- Business continuity planning during transitions

It is important to consider short-term mitigation strategies when it comes to legacy systems.

Holistically, when legacy information systems are present, a heavier burden is placed on a small number of people to ensure that these systems maintain compliance. Also, it is less likely that there are strategies in place for continual improvement throughout the system lifecycle.

3 Technology Benefits, Challenges, and Opportunities

The pharmaceutical industry is undergoing an era of significant transformation, driven by the emergence of Pharma 4.0 and marked by the integration of advanced technologies such as AI, ML, IoT, edge computing, automation, and data analytics. These digital advancements have the potential to revolutionize the end-to-end drug lifecycle, from patient demands, drug discovery, development, manufacturing, operations, quality, and distribution, to patient support. However, alongside the numerous benefits these emerging technologies offer, there are critical challenges that organizations need to address to fully harness their potential. This chapter highlights key benefits and challenges and proposes steps to help organizations align emerging technology implementation with their specific business and compliance requirements.

3.1 Key Benefits of Emerging Technologies

Key benefits of implementing emerging technologies in the pharmaceutical industry include, but are not limited to:

- **Improved Patient Outcomes:** These technologies help allow faster access to more effective and safer therapies, through innovative discovery, development, and execution solutions.

- **Reduced Regulatory and Compliance Issues:** Improvements in data security, data integrity, and auditability for systems and processes help reduce potential bottlenecks. Digital regulatory intelligence tools can also help keep track of regulatory updates.

- **Enhanced Agility and Collaboration:** Advancements in sharing digital data and information across the pharmaceutical value chain help improve overall agility and cooperation between key business partners.

- **Enhanced Efficiency and Productivity:** AI and automation streamline drug discovery, development, and manufacturing, accelerating candidate identification, optimizing clinical trials, reducing errors, and cutting operational costs.

- **Personalized Medicine:** Big data integration from genomics and health records enables tailored therapies, enabling improved efficacy, reduced adverse reactions, and enhanced patient outcomes.

- **Reliable and Robust Processes:** Predictive analytics and real-time monitoring ensure consistent quality, regulatory compliance, and patient safety, whereas IoT and advanced technologies prevent downtime and detect anomalies.

- **Cost-Effectiveness:** Automation and data integration minimize manual interventions, eliminate redundancies, improve scalability, reduce waste, and accelerate time to market, ultimately lowering costs for patients and healthcare systems.

3.2 Key Challenges of Transitioning to Emerging Technologies

Challenges faced by the pharmaceutical industry in transitioning to emerging technologies include, but are not limited to:

- **Data Integration and Cybersecurity Risks:** Managing vast amounts of data from IoT and digital tools is complex, requiring seamless integration, strong governance, and advanced analytics. Cybersecurity threats are heightened in cloud-based environments, necessitating robust protection frameworks.

- **Regulatory Compliance:** Regulatory frameworks struggle to keep pace with technological advancements, slowing the adoption of innovation. Organizations must navigate evolving guidelines on AI, automation, and continuous manufacturing while proactively engaging regulators to accelerate industry-wide adoption.

- **Legacy System Integration and Transition Costs:** Merging new technologies with legacy systems is resource-intensive, requiring seamless data traceability, compliance, and infrastructure upgrades. High costs for system updates, workforce training, and data management further add to the challenge.

- **Workforce Adaptation:** A skilled workforce proficient in digital tools, data analytics, and AI, alongside strong project management, collaboration, and communication skills, is essential for integrating emerging technologies. Bridging the gap between traditional pharmaceutical operations and new capabilities is key to unlocking their full potential.

  Continual upskilling and retraining are critical to scalability, preventing bottlenecks caused by a lack of specialized expertise. However, technical training alone is insufficient—successful transformation also requires strong change management leadership that drives cultural change, fosters continual learning, and encourages a mindset shift. Without this alignment, even the most capable workforce will struggle to maximize the benefits of emerging technologies.

- **Scalability:** As the pharmaceutical industry embraces emerging technologies such as AI, ML, gene editing, and personalized medicine, scalability becomes one of the major challenges. These innovations have the potential to revolutionize drug development and patient care, but transitioning from laboratory-scale breakthroughs to full-scale production requires significant complexity. A key scalability challenge is that the infrastructure required to support emerging technologies is often underdeveloped. Traditional manufacturing setups are not always compatible with the precision and customization demanded by advanced therapies. For instance, scaling gene therapies or personalized medicines necessitates customized manufacturing processes that differ significantly from the mass production of small molecules.

These shortcomings—compounded by the challenges of managing legacy systems, such as the difficulty of transporting, integrating, and making them accessible within new digital systems—necessitate the consideration of different models to keep pace with the evolving and emerging needs of the business. Refer to *ISPE GAMP Good Practice Guide: IT Infrastructure Control and Compliance (Second Edition)* [3] for specific details.

3.3 Strategic Alignment of Requirements for Digital Transformation

To accelerate and obtain support for digital implementation, organizations must be able to translate business and compliance needs into IT, OT, and other digital requirements. Driving the implementation of Pharma 4.0 requires a strategic approach to successfully integrate new technologies and digital solutions with existing systems, while providing incremental value and results using a phased development and implementation approach. The following steps are recommended to effectively translate these requirements into IT, OT, and digital specifications. By following these steps, organizations can effectively bridge the gap between business objectives and regulatory compliance, thereby enhancing operational efficiency and innovation in the pharmaceutical industry.

1. **Stakeholder Engagement:** Begin by engaging all relevant stakeholders, including business leaders, compliance officers, IT and OT teams, and end users. Understanding their needs, concerns, and visions for Pharma 4.0 will provide a solid foundation for the project. A multidisciplinary project execution team should be formed from this stakeholder group and can change over time based on project needs.

2. **Assessing Digital Maturity:** Establish a baseline to understand the organization's digital maturity. The *ISPE Baseline Guide: Pharma 4.0* [1] provides a framework for a Pharma 4.0 Maturity Assessment which can help the organization understand where it stands with respect to resources, information systems, organization, processes, culture, as well as data integrity. This baseline will help drive improvement and set the stage for the subsequent steps.

3. **Requirements Gathering:** Conduct comprehensive workshops or interviews to gather both business and compliance requirements. This should include understanding regulatory frameworks, quality standards, and operational needs unique to the pharmaceutical industry today and in the future.

4. **Gap Analysis and Risk Assessment:** Perform a gap analysis to identify discrepancies and key differences between current capabilities and future goals. This can highlight the root causes and where new technologies or processes are necessary to address these discrepancies. Identify and assess the risks associated with these gaps to support project prioritization.

5. **Defining IT and OT Specifications:** Translate business and compliance requirements into specific IT and OT specifications. This may involve selecting appropriate technologies, platforms, and data architectures that facilitate seamless communication between systems. Adopting standards, such as ISA-95 [18] for IT/OT integration and Open Platform Communications Unified Architecture (OPC UA) [19] for secure data exchange, ensures robust and scalable architectures.

6. **Latest Technologies and Tools:** Implement process mapping and other available tools to help visualize and simulate operations. This helps with optimizing processes and ensuring that new technologies align with key business and operational needs. Assess the viability of using emerging technological advancements, such as Digital Twins, Large Language Models (LLMs), and generative AI as well as agentic AI, among others.

7. **Regulatory Compliance Integration:** Incorporate compliance aspects, such as data integrity, audit trails, and traceability, into all IT and OT specifications to streamline regulatory approvals.

8. **Prototyping and Testing:** Build prototypes or pilot projects to test the viability of the proposed IT and OT solutions. This step provides a feedback loop to refine the specifications based on actual user experience and requirements.

9. **Operational Readiness and Preparedness:** Evaluate the current state, plan for the future state, and ensure all stakeholders are equipped to handle the changes.

10. **Training and Change Management:** Develop a comprehensive training plan for end users and implement change management strategies to ensure a smooth transition to the new technologies.

11. **Launch New Solutions:** Once systems, processes, and resources are fully prepared for the transition, activate the new solutions and officially go live with the operators, process and business owners, and supporting functions.

12. **Continual Improvement Framework:** Establish a framework for continuous monitoring and feedback once the systems are implemented. This can help adapt and optimize solutions based on evolving business needs and compliance landscapes.

13. **Information and Knowledge Sharing:** Capture all processes, specifications, and workflows accurately, leveraging digital tools to store organizational knowledge for sharing across the organization. Knowledge sharing across teams, combined with effective cultural change management, fosters a culture of collaboration and innovation in the ongoing digital transformation journey of implementing Pharma 4.0.

4 Regulatory Compliance

As the pharmaceutical industry advances into the Pharma 4.0 era, regulatory compliance hinges increasingly on robust data governance and a proactive approach to digital transformation. Health authority inspectors expect clear evidence of how data supports quality, safety, and decision-making, as required by regulatory guidelines. By establishing comprehensive data governance frameworks—centered on data integrity, security, traceability, and lifecycle management—organizations can meet these expectations with confidence. Additionally, process validation, process monitoring and control, and continuous process verification, including real-time release testing and parametric release, play critical roles in ensuring product quality and efficiency. Together, these strategies enable manufacturers to verify and control process variations, enhancing both compliance and production reliability. Furthermore, ongoing compliance requires cross-functional collaboration, continuous training, and alignment with evolving regulatory standards. Through diligent implementation of best practices, effective communication with regulators, and a culture of continual improvement, organizations can ensure both regulatory success and long-term operational excellence.

Leveraging data governance and frameworks is essential to fostering health authority inspectors' understanding of data-driven decisions. In the pharmaceutical industry, as in many industries, data-driven decision-making is crucial for ensuring quality, safety, efficacy, and regulatory compliance. As the industry evolves with Pharma 4.0 and embraces digital technologies, regulators increasingly focus on how organizations manage and use data, especially through Critical Quality Attributes (CQAs), Critical Process Parameters (CPPs), and Critical Material Attributes (CMAs). Health authority inspections, particularly under PIC/S [9], require a clear demonstration of how data governance and robust frameworks support decision-making (e.g., PI 041-1 [10]). Effectively understanding and communicating this to inspectors is vital for successful inspection outcomes. Additionally, proactively establishing robust data governance and frameworks significantly supports efficiency, quality, safety, efficacy, and regulatory compliance.

4.1 Key Components of a Data Governance Framework for Regulatory Compliance

- **Data Integrity and QA:** The data integrity principles of ALCOA++ are required for data over its lifecycle.
- **Establishing Robust Procedures:** Robust procedures help prevent data tampering, ensuring proper data entry practices. Regularly auditing data systems is crucial. A well-defined framework includes SOPs for maintaining data integrity, backed by training programs to ensure employee compliance and good understanding and execution.
- **Data Access and Security:** Ensuring that only authorized personnel can access sensitive data is a critical element of any data governance framework. This is particularly important for health authority inspectors, who require assurance that data access is tightly controlled and systems are equipped to prevent unauthorized access or data breaches. This can be demonstrated through role-based access controls, audit trails, and periodic reviews. Additionally, data classification and inventory ensure the right controls are applied adequately. A robust data incident response and breach notification procedure aids in identifying and resolving issues promptly. Lastly, encryption measures should align with 21 CFR Part 11 [11], EU EudraLex Volume 4 Annex 11 [12], and PIC/S Annex 11 [20] requirements on computerized systems.
- **Traceability and Documentation:** One of the key principles of data governance is traceability. Inspectors expect that any data used in decision-making can be traced back to its origin, with a clear and documented history of any changes. This means that developers must have a robust and transparent understanding of the algorithms, procedures, and neural networks used in the decision-making process. Organizations also need to implement systems that automatically record all actions performed on data, ensuring that inspectors can verify the authenticity and reliability of the data.

- **Data Lifecycle Management:** Managing the lifecycle of data—from creation and processing to archiving and disposal—is crucial. It is important to note that the data lifecycle includes data generated from the process lifecycle and the product lifecycle. A comprehensive data governance framework ensures that data retention policies are in line with regulatory requirements and that data disposal is handled securely. Ensuring that data remains retrievable and readable throughout its lifecycle helps demonstrate compliance with PIC/S guidelines [9].

- **Integration of Data from Multiple Sources:** Data often comes from a variety of sources, including clinical trials, manufacturing processes, Quality Control (QC), analytical instrumentation, data systems, historians, and supply chain operations. A well-structured data governance framework integrates these disparate data sources into a system that allows for comprehensive data analysis. By following FAIR (Findable, Accessible, Interoperable, and Reusable) data principles [21], manufacturers can ensure data is available for all end users while maintaining access control. It should be demonstrable to inspectors that this integration supports decision-making processes that enhance product quality and patient safety.

- **Cyber Resilience:** Adherence to the Cyber Resilience Act (CRA) [22] is essential for robust security practices. The NIS2 Directive [23] strengthens cybersecurity by establishing EU standards, while the requirements in CRA ensure cybersecurity is embedded into digital products from the start. Aligning data governance strategies with these frameworks enhances proactive risk management and ensures compliance with evolving regulations, safeguarding digital assets effectively.

4.2 Demonstrating Data-Driven Decisions to Inspectors

Successfully navigating a health authority inspection requires not only having robust data governance practices in place but also effectively communicating these practices to inspectors. Organizations should be prepared to present their data governance framework in a way that clearly demonstrates how data is used to inform critical decisions. This involves the following:

- **Clear Digital Documentation:** All data governance policies, procedures, and frameworks must be well-established, well-documented, and easily accessible to inspectors. Documentation should provide a clear explanation of how data is governed, including the controls in place to ensure data quality and integrity.

- **Real-World Application:** Be ready to showcase real examples of how data governance has influenced decision-making. Whether it is through a use case or using an example of electronic Batch Records (eBR) systems, demonstrating the practical application of data governance is key to gaining inspector confidence. To showcase real examples with regulators, it is imperative to create a strategic plan on how to best share the electronic version of the data and avoid printing records.

- **Interactive System Demonstrations:** Be prepared to provide live demonstrations of data management systems to inspectors upon request. This can provide a tangible look at how data is collected, validated, and used in real time to make decisions. Highlight features such as automated audit trails, real-time data analytics, and secure access controls that align with PIC/S [9] expectations.

- **Training and Competence:** Ensure that employees are well-trained, competent in data governance and data integrity, and able to answer inspector questions confidently. Training records should be available to demonstrate that staff are competent in data governance practices and understand the regulatory implications.

- **Leadership Commitment:** Senior management oversight is integral to effective data governance practices. Through regular reviews—whether monthly, quarterly, bi-annual, or annual—such as a digital steering committee or quality management review, leadership demonstrates a strong commitment to understanding and maintaining robust data processes. This proactive engagement not only ensures compliance and continual improvement but also provides substantial confidence to regulatory inspectors, showcasing an organizational culture dedicated to upholding the highest standards of product quality and patient safety.

4.3 Aligning Data Governance with PIC/S Guidelines

The PIC/S guidelines [9] serve as a benchmark for GMP inspections worldwide, and aligning the data governance framework with these guidelines is essential. Key PIC/S guidance related to data includes:

- **PIC/S Annex 11 [20]:** Focuses on computerized systems and their validation, emphasizing data integrity, security, and traceability.

- **PIC/S Annex 15 [24]:** Covers the qualification and validation of manufacturing processes and systems, including the role of data in ensuring consistent quality.

- **PIC/S Guide to Good Practices for the Preparation of Medicinal Products in Healthcare Establishments – Chapter 4 [25]:** Stresses the importance of data integrity in the distribution and storage of pharmaceutical products.

Alignment of the data governance framework with these PIC/S guidelines can demonstrate to inspectors that the organization not only complies with regulatory expectations but also proactively manages data to improve decision-making, product quality, and patient safety.

To summarize, the Pharma 4.0 era emphasizes data centricity in decision-making, requiring robust data governance frameworks for regulatory compliance. For health authority inspectors, especially under PIC/S guidelines [9], understanding how data governance supports decision-making is vital. By ensuring data integrity, traceability, security, and regulatory alignment, pharmaceutical organizations can build confidence with inspectors, demonstrating that their data-driven decisions uphold product quality and patient safety as the ultimate goal.

4.4 Areas of Focus

Ensuring compliance with regulatory requirements during the Pharma 4.0 transformation is critical for maintaining product quality, patient safety, and operational integrity. It is important to emphasize that this is not a one-time project, but rather an ongoing process. By focusing on these recommended practices, organizations can enhance their ability to comply with regulatory requirements during Pharma 4.0 implementation, ultimately ensuring product safety and efficacy. Additionally, ensuring compliance and developing processes as compliant by design avoid delays to market and interruptions in production. This ultimately leads to cost reductions, accelerated time to market, and more predictable business and manufacturing.

Recommended areas of focus and good practices are as follows:

- **Understand Regulatory Frameworks:** Maintain familiarity with relevant regulations and guidelines (e.g., FDA [26], EMA [27], ICH [28]) and industry standards for pharmaceuticals. This includes understanding how these regulations apply to digital technologies and data management.

- **Early Engagement with Regulatory Affairs:** Involve regulatory compliance, regulatory intelligence, IT, data, and quality teams from the beginning of the implementation process and throughout the technology lifecycle. Their insights can help shape strategies that align with compliance requirements.

- **Risk Management:** Conduct comprehensive risk assessments to identify potential compliance risks associated with new technologies or processes. Implement risk mitigation strategies to address these concerns proactively.

- **Data Integrity and Security:** Ensure that data integrity is maintained throughout the data lifecycle. Implement robust data management practices, including validation protocols, to ensure the accuracy and reliability of data, while securing data and establishing adequate access control systems.

- **Documentation Practices:** Maintain thorough documentation of processes, systems, and changes. This includes SOPs, validation protocols, and change controls to ensure traceability and accountability.

- **Validation of Systems and Processes:** As regulatory standards evolve, using effective industry best practices and guidance on AI and ML is crucial for implementing robust validation protocols for software, hardware, and processes. This ensures systems not only meet but potentially exceed regulatory standards, affirming their suitability for intended use and compliance with GMP. By adopting these practices, organizations can enhance their operational excellence and adeptly navigate the complexities of integrating advanced technologies.

- **Training and Competency:** Provide training for employees on compliance requirements and the proper use of new technologies and systems. Regularly assess employee competency to ensure adherence to the latest regulatory standards.

- **Change Control Procedures:** Establish robust change control processes to manage modifications to systems, processes, or equipment. This ensures that all changes are evaluated for compliance impact before implementation.

- **Audit and Compliance Checks:** Conduct regular internal audits and compliance checks to assess adherence to regulatory requirements. This helps identify areas for improvement and ensures ongoing compliance.

- **Data Privacy and Protection:** Adhere to data privacy regulations (e.g., EU General Data Protection Regulation (GDPR) [29], US Health Insurance Portability and Accountability Act (HIPAA) [30]) when handling sensitive patient or research data. Implement privacy by design principles for commercial off-the-shelf, configurable, and customizable solutions to ensure data privacy aspects are applied to all digital solutions. Use measures to protect personal information and ensure data is only used for its intended purpose.

- **Collaboration with IT and Quality Teams:** Foster strong collaboration between IT, QA and regulatory teams to ensure that technology implementations align with quality standards and regulatory expectations.

- **Industry Trends:** Monitor and stay informed about evolving regulatory landscapes and industry best practices. This can help organizations anticipate changes and adapt their compliance strategies accordingly.

- **Engagement with Regulatory Authorities:** Maintain open communication with regulatory authorities. Engage them during the development of new technologies or processes to gain insights and feedback. This can be done using dialogue platforms offered by industry associations.

- **Vendor Selection:** Another important consideration is selecting vendors that are knowledgeable about pharmaceutical regulations and thus can provide a reliable partnership. These vendors should understand how their solutions integrate with other systems and any associated risk implications.

- **Post-Implementation Review:** Conduct a thorough review after implementation to assess compliance and identify lessons learned. This can help inform future projects and continual improvement efforts.

- **Continuous Monitoring:** Continuous monitoring is a key driver of both regulatory compliance and process efficiency in Pharma 4.0. By leveraging real-time data and automation, it enables early detection of deviations, ensures consistent product quality, and strengthens data integrity and traceability—supporting faster, safer product release and regulatory readiness. Simultaneously, it enhances processes through real-time adjustments, predictive maintenance, and data-driven optimization, reducing downtime, waste, and batch failures. This makes manufacturing more agile, stable, and cost-effective.

5 Data Governance

A comprehensive and robust data governance framework creates the structure required to manage data as a critical asset. The combination of process and technical controls allows organizations to apply flexibility where needed when managing data across diverse application landscapes and complex infrastructure. It is only through diligent oversight, strategic evolution, and strict adherence to this framework that organizations can truly unlock the value of the data they possess, driving continual improvement, and the ability to pursue emerging technologies.

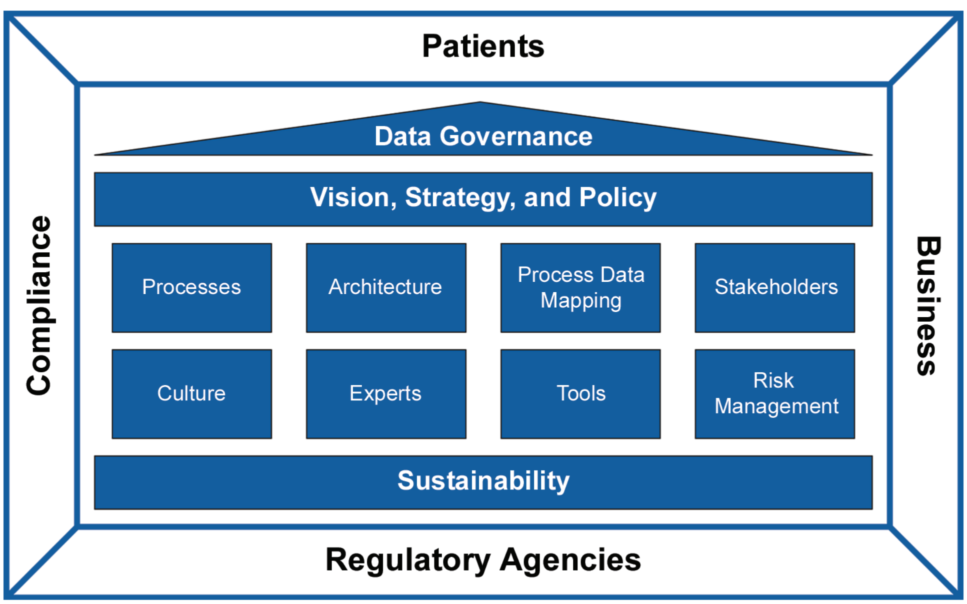
5.1 Data Governance Foundation

Data governance is a critical element of any digital transformation. It ensures the ability to successfully identify and implement process improvements while also establishing a foundation for the delivery of emerging technologies. In the pharmaceutical industry, data governance is a strategic framework for ensuring data integrity, security, and accessibility throughout its lifecycle. It safeguards data as a valuable asset, supports regulatory compliance, and aligns with the organization's mission. A robust governance system optimizes data usage while mitigating risks such as breaches and misuse.

A data governance system should account for both the technical aspects of data management and stewardship as defined within the FAIR principles, but also the more end-user-facing guiding data integrity elements as defined within the ALCOA++ framework. Inclusion of both sets of principles ensures the creation of a robust data governance framework, capable of supporting current organizational needs and strategic goals and objectives.

A data governance program establishes a structured approach to managing data assets through clear processes, policies, and practices. It ensures data quality, integrity, security, and compliance while promoting cross-functional collaboration. The roadmap of data governance consists of the following elements as illustrated in Figure 5.1 and explained in the following sections.

Figure 5.1: End-to-End Data Governance



Data is the backbone of business operations, patient care, and regulatory compliance. A strong data governance framework is essential to managing, protecting, and utilizing data effectively. Think of data as a house: without a solid foundation and structured framework, it becomes difficult to maintain security, accessibility, and integrity.

A clearly defined and articulated data governance strategy ensures that data is secure, accessible, and reliable, which fosters transparency, collaboration, and informed decision-making. It also strengthens compliance with regulatory requirements such as PIC/S [9] and GDPR [29], safeguarding sensitive information, and reinforcing trust with regulators, patients, and business partners. It is important to ensure that the strategy allows for the flexibility and scalability required by the organization, understanding the current data/digital maturity as well as any potential challenges and limitations to adoption.

The benefits of a robust data governance framework include the following:

- Improved collaboration across departments, aligning data management with corporate goals
- Increased visibility and control over data governance activities
- Enhanced transparency, ensuring accountability in decision-making
- Effective monitoring of data usage and compliance with regulatory standards

By establishing clear policies and best practices, organizations can create a data-driven culture that supports innovation, regulatory adherence, and long-term success.

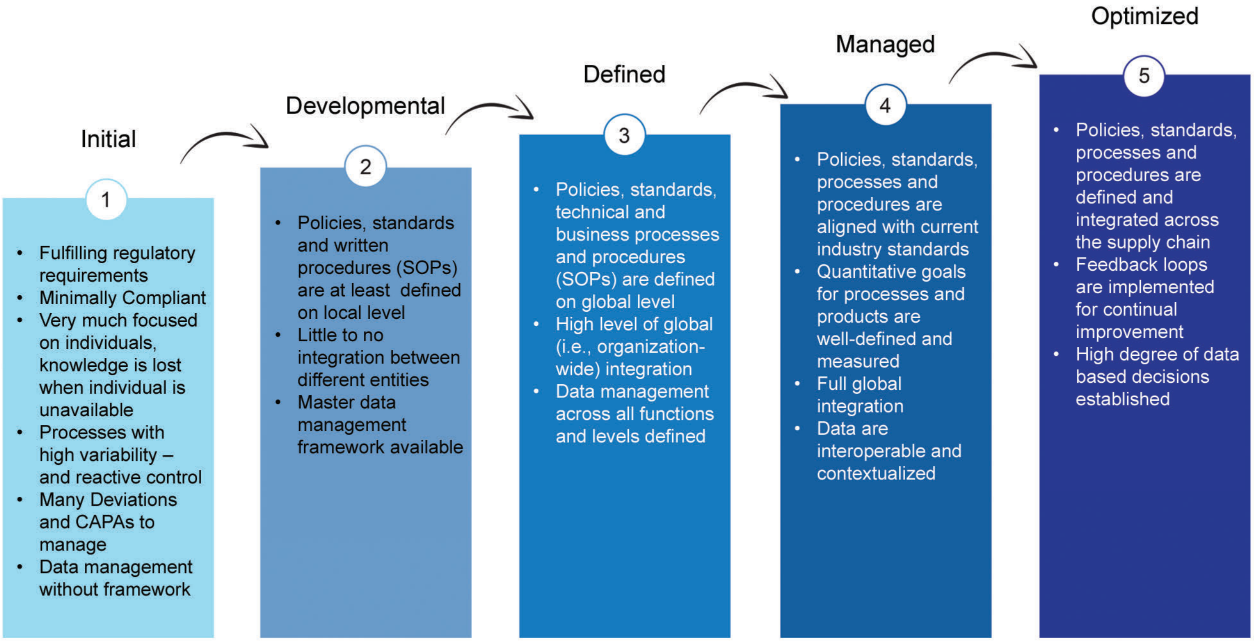
5.2 Characteristics of Good End-to-End Data

When considering the quality and "health" of organizational data, it is important to take a broad and holistic view. Consideration should be given across the entire solution and application landscape, and across all solution environments (e.g., development, validation, production).

Appropriate oversight, management, and control from end-to-end, enables reliable and predictable performance through the Software Development Lifecycle (SDLC) and resulting solution delivery. Data management principles and framework should consider the following as practical steps in the promotion of data stewardship, remaining aligned with FAIR and ALCOA++, as keys to enabling a successful and sustainable digital transformation:

- Leverage data stewards and product and process owners to ensure Master Data Management (MDM) accuracy and consistency within source systems. To the extent possible, master data should be harmonized and standardized across the organization, not unique or localized at a department or site/plant level.
- Drive the ingestion of selected data elements necessary to enable quality-related decisions into appropriate data repositories (i.e., data warehouses, data lakes) enabling downstream utilization. Ensure that all data relevant for a quality-related decision can be used by the decision makers.
- Provide controlled access to replicated source data within the data repositories (as opposed to source systems), facilitating data discovery, data modeling, Self-Service Business Intelligence (SSBI), and advanced analytics without impacting transactional system performance.
- Turn data into information, unlocking, demonstrating, and articulating business value via dashboards, visualizations, and Key Performance Indicators (KPIs) through enterprise reporting and analytics tools.
- Upskill team members through targeted and tailored training, enabling data-informed decisions, advancing overall digital maturity (see Figure 5.2), and building a future-fit workforce.

Figure 5.2: Maturity Level Definitions [1]



As a vital corporate asset, data should be seen as a strategic enabler, going beyond compliance requirements to drive innovation, decision-making, and competitive advantage. Thus, a strong data foundation is critical to the success of any digital transformation, including entry into the ever-evolving AI/ML space. Data has evolved to be considered a “product”, and must be continually monitored and managed, allowing team members to keep pace with the demands of customers and all stakeholders, from top management through the entire organization. Prioritizing the management of data can then drive a critical and necessary shift from data as a liability to data as an asset, allowing for a more impactful and transformational change.

Establishing and maintaining strong partnerships with system and process owners, data stewards, and functional support teams sets the foundation for a robust data management and governance framework, ensuring the necessary data can meet the expectations of customers and business stakeholders. The data landscape should be periodically reviewed and evaluated at points throughout the life of the solution to ensure it remains relevant, representative, and compliant.

- **Relevant:** Is the data inclusive of all elements required to support the evolving process or initiative? Have too many points been included (via metrics and KPIs) and introduced “noise”, or have critical parameters been excluded and created an incomplete picture?

- **Representative:** Is the data being utilized representative of the process and does it support the desired end state? Does the data accurately reflect the process and support the intended outcome? Testing and validation across all operational environments (e.g., development, validation, production) should include diverse data sets and not only ideal scenarios to ensure success in production, ensuring both positive and negative testing scenarios are considered to allow for successful production use.

- **Compliant:** Does the data continue to remain in a secure state, free of legal, regulatory, privacy, ethical, and bias concerns? When feasible and where required, data should be anonymized to ensure compliance with all applicable security and privacy concerns. Engage functional SMEs to ensure a state of control is maintained.

5.3 External Influencing Factors

There are key external factors that shape the structure of data governance in the pharmaceutical industry:

- **Health Authority Regulations:** Regulatory bodies such as the FDA [26], EMA [27], and WHO [31] set strict guidelines for data management to ensure that all pharmaceutical products meet safety and efficacy standards.
- **Patients' Unmet Needs:** Data governance enables organizations to respond to patient needs by ensuring accurate and appropriate access to data (aligned with applicable authorized use requirements) for research, clinical trials, and drug development.
- **Privacy Regulations:** Ensuring compliance with regulations such as GDPR [29] or HIPAA [30] is fundamental to maintaining patient trust, ensuring data privacy, and avoiding costly penalties.
- **Enhanced Cybersecurity:** Emerging threats necessitate the application of new and/or additional controls to ensure the secure management, transmission, and use of data.
- **Innovation/Emerging Technology:** Rapid evolution of emerging technologies, such as AI, requires continual evaluation and assessment of how solutions are governed.

The assessment of these factors throughout the lifecycle of data governance is shown in Table 5.1.

Table 5.1: External Factors Assessment Throughout the Data Governance Lifecycle

Lifecycle Stage	Health Authority Regulations	Patients' Unmet Needs	Business Success	Compliance Requirements
Data Creation and Collection	Ensure data complies with FDA, EMA, and WHO standards from the outset.	Collect patient-centric data to address treatment gaps.	Capture accurate, relevant data for strategic decisions.	Embed privacy controls (GDPR, HIPAA) at data entry.
Data Storage and Protection	Secure storage to meet data integrity and retention mandates.	Ensure data is accessible for patient-focused research.	Optimize storage for accessibility and scalability.	Use encryption and access controls to protect data.
Data Usage and Sharing	Share data responsibly for regulatory submissions and audits.	Enable data sharing to accelerate research and trials.	Promote cross-functional access to drive innovation.	Control sharing with consent management and compliance tracking.
Data Archiving and Retention	Follow retention timelines and secure archiving practices.	Preserve data for long-term patient outcomes monitoring.	Retain historical data to guide future strategies.	Align archiving with legal and regulatory mandates.
Data Disposal and Destruction	Dispose of data according to regulatory standards.	Ensure patient data privacy with secure disposal.	Manage lifecycle costs by responsibly removing obsolete data.	Verify destruction to comply with GDPR and similar laws.

5.4 Strategy and Policies

A data governance strategy must align with the organization's overall goals and objectives. Defining a clear vision for how data should be managed, protected, and utilized is crucial for setting the foundation of the governance framework. The program begins by developing comprehensive policies that guide data management, usage, security, and compliance. These policies should be developed in collaboration with stakeholders from various departments and aligned with the organization's strategic goals and priorities, ensuring that all data-related activities adhere to established organizational standards. Policies for data retention, archiving, and disposal should be established to manage data effectively over time. Policies should focus on the following:

- **Data Ownership and Accountability:** A key component of data accountability is assigning clear ownership of data to individual data stewards or teams responsible for maintaining its quality. This is to ensure compliance with regulations and policies, data quality is maintained, and access controls are managed. This accountability structure helps ensure that data is consistently managed and protected.

- **Data Standards and Metadata Management:** To achieve consistency and interoperability, the program establishes data standards across formats, naming conventions, and definitions. Metadata management processes document data lineage, definitions, and usage policies, ensuring that all users understand how data is collected, transformed and applied.

- **Data Collection and Storage:** Identifying what data is collected and stored is essential for understanding its significance, relevance to the organization, and impact on decision-making processes.

- **Data Usage and Access:** Establishing protocols for how data is accessed and used ensures that it is handled appropriately, ethically, and in line with organizational goals and regulatory requirements.

- **Data Accuracy and Reliability:** Implementing validation mechanisms and consistency checks ensures that data remains accurate, reliable, traceable, and up to date, supporting decision-making processes and operational efficiency.

- **Data Quality Management:** Ensuring high data quality is a priority by establishing, for example, a data governance program. A data governance program implements tools, metrics, and processes to identify, monitor, and correct data quality issues. This ensures that data remains accurate and fit for business use.

- **Data Security and Compliance:** Safeguarding data though robust security measures is essential to protect it from unauthorized access, breaches, and misuse while ensuring regulatory compliance and maintaining trust in the organization's data practices. This includes implementing controls and audits to enforce data protection and industry compliance.

6 Data Governance Core Elements

A well-structured data governance framework ensures consistent, secure, and compliant data management, transforming data into a strategic asset that enhances operational efficiency, accountability, and regulatory adherence. For the pharmaceutical industry, investing in data governance is essential to maintaining high data quality, security, and accessibility throughout its lifecycle. Clear policies, standards, and processes foster collaboration, informed decision-making, and compliance with evolving regulations, reducing risks while aligning with strategic business objectives.

6.1 Data Architecture

The successful implementation of Pharma 4.0 relies on a strong architecture by adopting key data standards, systems, and formats that facilitate interoperability, data integrity, and compliance with regulations. Some of the important components are covered in the following sections.

6.1.1 Data Standards

Industry standards are extremely important for enabling the implementation of Pharma 4.0, as a lack of standards results in customized formats, data sets, and connectivity methods. Standards provide a guideline for the industry to create digital systems and processes that are shareable with all partners, more flexible and scalable, and easier to maintain. There are various important data standards that cover many of the different product lifecycle stages and processes within the pharmaceutical industry. The following list identifies some of the major standards to consider, along with brief descriptions:

- Clinical Development and Patient Data
  - CDISC (Clinical Data Interchange Standards Consortium) [32]: Critical for clinical research, providing data standards for clinical trials, including Study Data Tabulation Model (SDTM) and Analysis Data Model (ADaM).
  - HL7 (Health Level 7) [33]: Standards for the exchange of information related to health care; useful for interoperability between systems in clinical settings.
  - FHIR (Fast Healthcare Interoperability Resources) [33]: A standard developed by HL7 that facilitates the exchange of healthcare information digitally.
- Manufacturing and Process Control
  - ASTM Standards [34]: International standards organization with a sub-committee for the pharmaceutical manufacturing industry. A key example is ASTM E3077-17 which covers eData exchange between suppliers and manufacturers.
  - OPC UA [19]: A platform-independent standard for secure and reliable data exchange in industrial automation, enabling interoperability between machines and systems.
  - ISA-95 [18]: A standard that bridges the gap between Enterprise Resource Planning (ERP) and manufacturing (MES) systems, providing models and terminology for seamless integration and data consistency.
  - ISA/IEC 62443 [35]: A cybersecurity framework for industrial automation systems, offering guidelines to protect against cyber threats and ensure secure operations across industrial networks.

- RAMI 4.0 (Reference Architectural Model for Industry 4.0) [36]: A framework that structures Industry 4.0 concepts, supporting digital twin integration and aligning IT and OT systems for smart manufacturing.

• Quality Systems

- *ISPE GAMP 5 (Second Edition)* [2] for validation and risk-based lifecycle management of computerized systems and data. Also refer to the *ISPE Baseline® Guide: Commissioning and Qualification (Second Edition)* [37].

- ICH Quality Guidelines:

  > Q8(R2) Pharmaceutical Development [38]

  > Q9(R1) Quality Risk Management [15]

  > Q10 Pharmaceutical Quality Systems [8]

• Other

- ISO (International Organization for Standardization) [39]: Standards applicable to pharmaceuticals range from different management systems such as ISO 9001 for quality management or ISO 27001 for information security management systems to data quality and master data in ISO 8000.

- BIM (Building Information Modeling): A standard method for digital information management in construction, allowing for a full digital interface between the construction and operations.

- Ontology standards: Standard defining terms to ensure common usage. An example is Allotrope Foundation Ontologies [40].

6.1.2 Data Sources and Systems

There are many sources of data generation that are important to capture, in order to better understand and manage key processes used throughout the product lifecycle. Data is generated by humans, sensors, or instruments and captured in various systems. Some of these key systems are noted next with a short description.

• IT Data Sources

- MES: Tracks and documents the transformation of raw materials into finished products. MES supports real-time production monitoring and regulatory compliance.

- Laboratory Information Management System (LIMS): Manages samples, associated data, and laboratory workflows, ensuring accurate testing and compliance with quality standards.

- ERP system: Integrates core business processes; useful for managing supply chain, inventory, and compliance documentation.

- Clinical Data and Document Management System (CDDMS): Streamlines the collection, validation, and analysis of clinical trial data.

- Warehouse Management System (WMS): Manages and controls the day-to-day operations of a warehouse, distribution center, or fulfillment center.

- Learning Management System (LMS): Manages all data associated with training compliance, ensuring it can be demonstrated that employees are trained to perform their assigned tasks.

- Lifecycle Management System: Platforms used to manage the lifecycle of products, validation, processes, and applications.

- Quality Management System (QMS): Formalizes and streamlines the management of quality processes by documenting processes, procedures, and responsibilities; it serves as a central repository for data related to compliance, training, risk management, and performance metrics, thereby continually enhancing product quality and operational efficiency.

- Electronic Laboratory Notebook (ELN): A digital tool that replaces traditional paper laboratory notebooks for capturing and managing experimental data.

- OT Data Sources

  - SCADA: Monitors and controls industrial processes remotely by collecting real-time data from sensors and equipment.

  - DCS/PLC: Automates and controls industrial processes locally, ensuring precise and reliable operation of machinery and systems.

  - Process historian: Collects, stores, and retrieves time-series data from industrial processes for analysis, reporting, and optimization.

  - IoT sensors and devices: Capture real-time data from physical environments, enabling smarter monitoring, automation, and decision-making across connected systems.

  - Building Automation System: Automates and controls mechanical systems including HVAC and plant utilities.

- Other Data Sources

  - Open Government Data (OGD) platform: Databases and platforms to manage public and private data for federated reporting and research.

  - External reference data: Reputable sources offer reference data sets, to ensure a greater variety and to better train AI/ML.

6.1.3 Data Formats

Using standard data formats that are structured and machine-readable enables increased flexibility in the types of platforms and connections that different partners can use for efficient data sharing. Some of the most widely used and applied formats are as follows:

- XML (Extensible Markup Language): Widely used for data representation and exchange. Some structures are formally recognized such as FHIR by HL7 [33], CDISC [32], B2MML [41], and OPC UA (IEC 62541-5 [42]).

- JSON (JavaScript Object Notation): Often used in web applications for transmitting data. Structured protocols are Message Queuing Telemetry Transport (MQTT) with Sparkplug [43] or OPC UA PubSub JSON [19] which define semantic context and payload structure.

- CSV: Simple format for data import and export, commonly used for tabular data representation. Potential challenges with compliance and auditability due to lack of structure and metadata.

- EDM (Electronic Document Management): Portable Document Format Archival (PDF/A) for regulatory submissions, ensuring documents are in a standardized format suitable for long-term archiving. Ideally, embedded XML makes information electronically available.

6.1.4 Interoperability Standards

Another critical factor for driving the expansion of Pharma 4.0 implementations is the ability to operate between different systems and applications. There are a number of existing interoperability standards as described next, depending on the specific applications and use cases:

- Interoperability Standards Advisory: Guidance on health IT standards and technologies for interoperability in healthcare, including real-world evidence and patient data interoperability.

- SPIRIT (Standard Protocol Items: Recommendations for Interventional Trials) [44]: Ensures that clinical trial plans and results are communicated clearly and effectively.

6.2 Process Data Mapping

6.2.1 Process Maps to Support Information Handling

Pharma 4.0 initiatives come with an increase in process knowledge. These vast amounts of information need to address operators and systems alike. A solution to bridge the human and IT world of information handling lies in process maps. Process maps are deterministic model representations of real-life (business) processes, visualized as flowcharts. They include defined input, output, and the sequence of tasks and decisions (i.e., gateways) in between. They efficiently hold high amounts of structured information and metadata such as ownership, participant interfaces, and risk. By following a clearly defined syntax such as BPMN 2.0 (Business Process Management Notation 2.0, developed and maintained by a nongovernmental organization), they address stakeholders from operations and IT alike. However, transforming full text into process maps and storing them as pdfs on a share file is an insufficient "paper-on-glass" approach (i.e., where data is viewed on a glass screen instead of paper, but the information is not digitally usable).

Application of dedicated software for BPM allows effective documentation of process charts and associated metadata. This enables connected level structures until the required depth of documentation is reached. Cross-referencing and central repositories ensure that ALCOA++ principles and data governance are fulfilled. Depending on organization size, budget, and maturity, various BPM solutions are available. In addition, new cloud-based Software-as-a-Service (SaaS) platforms for BPM and automation continually emerge in this dynamic market.

Many organizations already focus on the documentation of end-to-end processes and an increased process orientation. These initiatives connect department silos and achieve efficiency gains along the transparent value chain. This approach can be extrapolated across multiple organizations to handle the increased complexity of modern ecosystems. Associated methods such as EAM manage information on individual IT applications and underlying data assets. Building and maintaining centralized inventories for processes and systems is key to achieving a holistic approach to digital enablement. Ideally an integrated approach is followed, contributing towards the concept of a DTO.

6.2.2 Adding a Data-Centric Perspective

As mentioned throughout this Guide, the relevance of "data" strongly increases in the Pharma 4.0 era. Organizations can benefit from moving from a process-centric viewpoint to a data-centric viewpoint. Based on process maps and data asset registers, a transformative data-centric view is promoted to foster critical thinking for identification of transformation potential along the entire process landscape:

1. Identify and map end-to-end processes to establish a process-centric view. Assign processes to adequate process owners. Start with selected processes (e.g., in the scope of a specific transformation project) and scale to the whole organization over time. The same scaling approach can be used for BPM applications.

2. Identify and register IT applications, including underlying data assets and their respective data asset owners. Again, this should start with an initial project and scale to an organization-wide approach, supported by appropriate EAM tools and platforms.

3. Map data assets to process tasks.

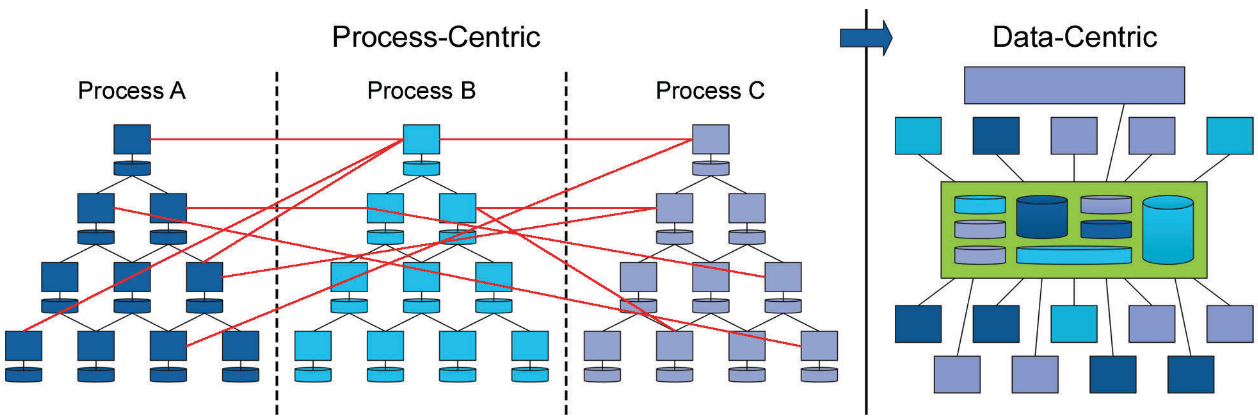
4. Map data assets in tasks to their respective phases in the data lifecycle (e.g., according to the *ISPE GAMP Guide: Records & Data Integrity* [14] and including the creation; processing; review, reporting and use; retention and retrieval; and destruction phases), respective storage format (e.g., database, paper, file share, or spreadsheet), and level of integration (e.g., according to the RAMI model: asset integration, communication, information, function, and business).

5. Reorganize the tasks from multiple processes where an individual data asset is involved according to its lifecycle stages. This transformation shifts the process-centric-view to a **data-centric view**, as shown in Figure 6.1. The result allows for a holistic assessment of an individual data asset along its lifecycle across multiple processes. It reveals potential inconsistencies of data asset handling such as duplications in creation, different storage formats, and undesired modifications.

6. Apply critical thinking to develop sound options for cross-process transformation, grounded on a holistic understanding of the underlying data assets to benefit efficiency, compliance, and data integrity.

7. Derive and execute actionable transformation projects as part for the wider Pharma 4.0 transformation program.

Figure 6.1: Changing the Perspective from the Process-Centric to Data-Centric Approach [45]



BPM and EAM combined with the right methodology such as the process data mapping approach are key elements to enable holistic business transformation and achieve transformation excellence.

6.3 Data Exchange Good Practices

Implementing Pharma 4.0 requires a structured approach to digital data exchange to ensure efficiency, compliance, and data integrity. The following good practices for digital data exchange support decision-making during design of target procedures in the transformation journey. By implementing these good practices, organizations can optimize digital data exchange processes within Pharma 4.0, leading to enhanced operational efficiency, regulatory compliance, and ultimately, better patient outcomes.

1. **Adopt Standardized Data Formats:** Use standardized data formats such as XML, JSON, or CSV in specified structures such as FHIR and MQTT to facilitate interoperability between different systems and applications. This helps ensure that data can be easily shared, interpreted, and utilized across various platforms.

2. **Implement Robust Data Governance:** Establish clear data governance policies that define data ownership, data stewardship, data quality standards, access controls, and data lifecycle management. This ensures accountability and consistency in data handling.

3. **Ensure Data Integrity and Compliance:** Use validation checks and audit trails to maintain end-to-end data integrity. Compliance with relevant health authorities' regulations needs to be incorporated into all data exchange processes, ensuring that records can be trusted and retrieved when necessary.

4. **Utilize APIs for Seamless Integration:** APIs should be employed to facilitate real-time data exchange between disparate systems (e.g., MES, LIMS, ERP). RESTful (Representational State Transfer) APIs allow for flexible integration with web services.

5. **Prioritize Cybersecurity and Data Security:** Implement strong cybersecurity measures, including encryption for data at rest and in transit, user authentication, and access controls. Data security is paramount, especially when dealing with sensitive patient—e.g., Protected Health Information (PHI), Personally Identifiable Information (PII)—or proprietary information such as Intellectual Property (IP).

6. **Promote Interoperability:** Advocate for interoperability standards such as HL7, FHIR, or ISA for healthcare and life science industry digital data exchanges. These standards enhance the ability of different systems to communicate and share information effectively.

7. **Facilitate Training and Change Management:** Provide comprehensive training on digital tools and data exchange protocols to process owners, operators, and end users. Change management processes should be established to help teams adapt to new technologies and workflows.

8. **Establish a Data Exchange Framework:** Create a clear framework for data exchange that outlines protocols, formats, and procedures for data sharing. This framework should be consistently reviewed and updated to account for emerging technologies and regulatory changes.

9. **Leverage Cloud-based Solutions:** Consider cloud platforms for enhanced scalability, flexibility, and accessibility of data. Cloud solutions often come equipped with tools for secure data sharing and collaboration.

10. **Implement Version Control:** Use adequate systems to enable version control for documents and data sets to ensure that all stakeholders work from the most current and approved versions. This practice can prevent confusion and inconsistencies in data availability.

11. **Regularly Monitor and Audit Data Exchanges:** Establish processes for regularly monitoring and auditing data exchanges to identify potential issues or areas for improvement. This could involve automated monitoring tools that track data integrity and flows.

12. **Integrate Feedback Loops and Continual Improvement:** Incorporate feedback mechanisms to collect user experiences and insights regarding data exchanges. Use this information to refine and optimize processes and technologies continually. AI can also be leveraged for monitoring and alignment of master data, especially in dynamic organizations. This can be autonomous or with human oversight.

6.4 Building Blocks for Data Governance

Strong data governance requires various building blocks for an organization's unique data management strategy, serving as the foundation for organizing, controlling, and securing data assets. These building blocks provide a structured framework to ensure that data is reliable, accessible, and protected, ultimately improving decision-making, regulatory compliance, and operational efficiency. This section covers key building blocks. Their function within a governance roadmap is discussed in Chapter 8.

Key Functions

- **Data Organization and QC:** Automatically scans databases, detects errors, and ensures accuracy, consistency, and redundancy elimination.

- **Security and Compliance:** Manages access controls, tracks data lifecycle, and ensures compliance with regulations such as GDPR [29] to protect sensitive information.

- **Enhanced Decision-Making:** Delivers clean, reliable data for trend analysis, strategic optimization, and improved customer experiences.

- **Automation and AI Integration:** Uses AI-driven automation to streamline and scale governance processes.

- **Risk Management:** Supports ICH Q9(R1) [15] compliance by identifying, assessing, and mitigating risks, ensuring data integrity and security. Risk management is discussed further in Section 7.6.

- **Quality Management:** Supports ICH Q10 [8] compliance by establishing robust systems for continual improvement, ensuring product quality, regulatory adherence, and patient safety.

- **Metadata Management:** Manages the context, meaning, and structure of data to improve data discovery and usability.

- **Data Stewardship:** Assigns responsibility for data assets to ensure proper management, maintenance, and oversight.

How They Work

These building blocks integrate with existing systems, scanning on-premises and cloud-based data to classify sensitive information, enforce governance policies, and provide a comprehensive data overview.

Key Benefits

- Enhanced security with automated access control and encryption

- Improved efficiency through streamlined data processes

- Cost reduction by automating compliance and minimizing errors

- Superior data quality with real-time monitoring

When choosing a tool, organizations should assess ease of use, integration, scalability, and cost to align with their data governance goals.

6.5 Stakeholders and Roles for Data Governance

- **Executive Leadership:** Provides strategic direction and ensures the necessary support and resources for implementing data governance initiatives. They are responsible for defining high-level policies and ensures alignment with the organization’s goals.

- **System and Process Owners:** Develop and maintain the technical infrastructure necessary to support data governance efforts. In a no-code environment, IT professionals can focus on enabling systems that allow for easy implementation of governance policies while ensuring system security and compliance. Process, system, and data ownership may be closely connected and held by one or more aligned individuals.

- **Data Owners:** Responsible for the quality, integrity, and compliance of data within their assigned areas.

- **Data Stewards:** They are “hands-on” and typically members of the operational unit. They manage the quality, integrity, and compliance of data within their assigned areas to realize the governance policies.

- **Compliance Teams:** Oversee adherence to regulatory and internal standards related to data usage, security, and ethics in alignment with legal.

- **Data Scientists and Analysts:** Leverage governed data to derive insights that inform decision-making. Access to compliant and well-governed data allows analysts to generate accurate and timely reports, which are crucial for driving business outcomes.

- **Legal and Privacy Officers:** Oversee data privacy, consent management, and legal risk mitigation.

7 Data Governance Roadmap

In the Pharma 4.0 era, a strong data governance framework is essential for ensuring secure, compliant, and efficient data management. By transforming data into a strategic asset, organizations can enhance operational efficiency, accountability, and regulatory adherence. Investing in data governance is critical for maintaining high data quality, security, and accessibility throughout its lifecycle.

As an initial step, relevant stakeholders need to be identified and engaged. In parallel, the organization's culture towards data needs to be evaluated and developed. A unified data strategy ensures that no data is overlooked, improving governance effectiveness. Engaging prior identified stakeholders across the organization is crucial to shaping governance policies that align with business goals and regulatory requirements. Active data stewardship promotes collaboration, ensuring accountability and alignment with strategic objectives. Integrating governance practices into existing workflows, automating quality checks, and managing access controls fosters embedded QRM without disrupting operations. Continual training keeps employees up to date with governance principles and evolving regulations, while measurable KPIs track progress and drive improvement.

By setting up and executing their individual road map, pharmaceutical organizations can enhance compliance, mitigate risks, and sustain operational excellence. Data governance will not only ensure regulatory compliance but also empower organizations to thrive in an increasingly data-driven, digital landscape.

7.1 Identify and Engage Stakeholders

Identifying key stakeholders, as described in Chapter 6, and ensuring that they are literate in data governance principles is crucial. Proper training and continual awareness programs must be implemented to foster engagement and ensure that everyone understands their roles and responsibilities regarding data governance.

Data governance is a cross-functional initiative that requires collaboration among multiple stakeholders to ensure data is managed effectively, securely, and in compliance with regulatory requirements. The involvement of key stakeholders is essential to ensure that data governance is adaptable to evolving business needs and regulatory frameworks, such as FDA 21 CFR Part 11 [11] and PIC/S [9] requirements for computer system validation in the pharmaceutical industry.

The adoption of low-code or no-code environments can empower leadership to define the governance frameworks more efficiently, while IT professionals or data stewards can quickly implement and enforce them. This fosters a more collaborative approach across the organization, enabling seamless access to data for reporting and analysis while maintaining compliance and data quality.

7.2 Building a Data-Driven Culture

Once the organization is well-informed and stakeholders are engaged, the focus turns to creating a data-driven culture. This element emphasizes the importance of embedding data governance into the organizational DNA so that data becomes an integral part of every decision and process. Further details on people and culture are in Chapter 8.

7.3 Data Governance Design

Before implementing a data governance process, it is important to assess the scalability and robustness of the framework against changes in the organization. A proper data governance design is needed for this. It is recommended that the correct outlines of the various data sources have already been drawn. Accurate data architecture and possible data flow analysis must be available to be able to implement a well-thought-out data governance framework. This information can come from existing functional, configuration, and design specifications. The data governance framework and its core elements explained in Chapters 5 and 6 are the fundamentals for this design. Prior to data governance, a digital maturity assessment can also ensure that the design is in line with the business objectives.

7.4 Implementing Data Governance Processes

With stakeholder alignment, data-driven culture, and a well-defined data governance design, the next step is to implement robust data governance. Previously identified stakeholders ensure seamless integration of systems, tools, and processes across the organization. Thus, establishing clear data ownership, stewardship, and custodian roles are fundamental to digital transformation. Clear roles, responsibilities, and governance ensure accountability and consistency across data elements, models, systems, and processes. These processes ensure that data management is systematic, with controls for data integrity, accessibility, and security, and that they are consistently applied throughout the organization.

The program governs data throughout its lifecycle—from creation to archiving or disposal—based on business requirements and regulatory needs. Data governance has a direct link with process performance and product quality and is subject to proper management review. Key principles for sound data governance include the following:

- **Point of Contact (PoC):** Serves as the primary liaison for data-related matters within a function
- **Ownership and Decision-Making:** Defines data usage and policies within a department
- **Data Quality Management:** Ensures integrity, including MDM
- **Cross-Functional Coordination:** Engages the right resources for data-driven processes
- **Standardized Definitions and Policies:** Establish workflows, governance, and compliance protocols
- **Consistency and Accessibility:** Ensures uniform data usage and educates teams on data literacy
- **Monitoring and Compliance:** Regular auditing to align with internal and external regulations and developments
- **Strategic Guidance:** Advocates for data best practices in projects and initiatives

A well-defined data stewardship framework ensures alignment between owners and managers, clarifying the intended use and impact of data. Depending on the complexity and scope, multiple data owners and stewards may be required across functions and sites. Standardized systems, formats, and governance enhance interoperability, regulatory compliance, and Pharma 4.0 success, enabling seamless data integration across the pharmaceutical value chain.

7.5 Sustainability and Continual Improvement

Data governance is not a one-time initiative but rather an ongoing process. A sustainability plan ensures that the governance framework is maintained and updated as needed, adapting to changes in regulations or organizational needs. Celebrating successes and fostering a culture of continual improvement ensures that data governance evolves in step with the organization's growth.

7.6 Embedding Quality Risk Management

In the evolving pharmaceutical landscape, digital transformation, data analytics, and AI are reshaping drug development, manufacturing, and patient care. Central to this shift is robust data governance, ensuring data integrity, compliance, and reliability. QRM plays a critical role in navigating regulatory complexities and enabling data-driven decision-making to enhance efficiency, product quality, and patient safety.

7.6.1 The Role of Quality Risk Management in Risk Governance

QRM provides a structured approach to identifying, assessing, and mitigating data risks, as shown in Figure 7.1 from ICH Q9(R1) [15]. This approach involves the following steps:

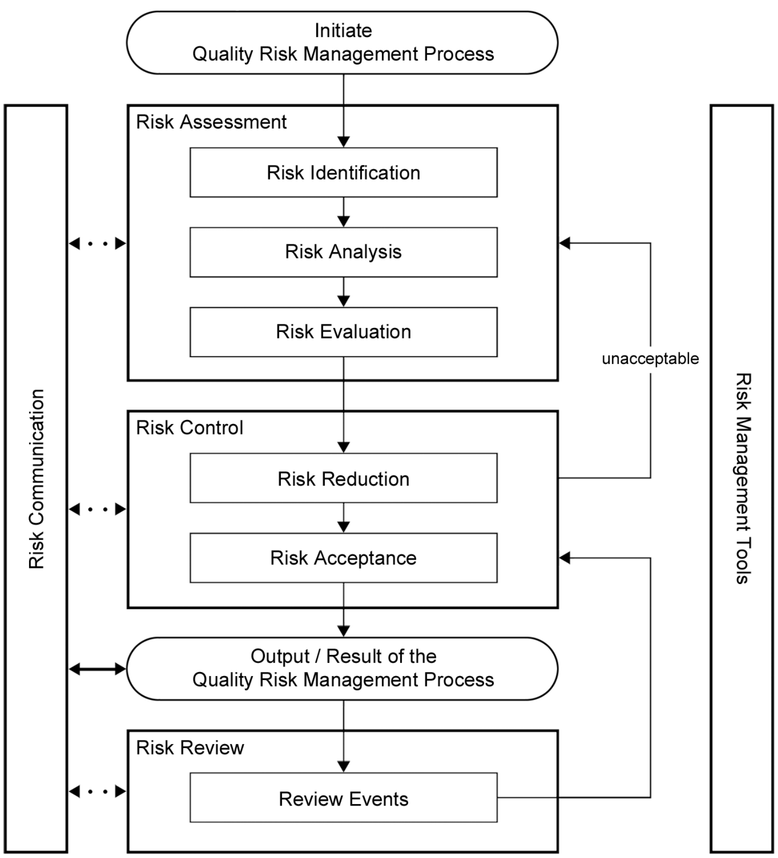
1. **Hazard Identification:** Identify important data and recognizes vulnerabilities in data collection, storage, and processing, including breaches, entry errors, and outdated data

2. **Risk Assessment and Prioritization:** Evaluate detectability, likelihood, and impact, focusing resources on high-priority threats

3. **Risk Control Implementation:** Introduce security protocols, data validation, and audits and improves the detectability to mitigate identified risks

4. **Continuous Monitoring and Review:** Ensure risk controls remain effective and can adapt to emerging threats

Figure 7.1: Quality Risk Management Process [15]



7.6.2 Enabling Data-Driven Decision-Making through QRM

Integrating QRM into data governance enhances the following:

- **Operational Efficiency and Smarter Investments in Data Infrastructure:** QRM encourages critical thinking and risk prioritization, helping organizations to reach the correct allocation of resources and to focus on data systems and processes that are most critical to product quality and patient safety.

- **Strengthening Data Integrity:** QRM helps identify and mitigate risks that could compromise data integrity.

- **Regulatory Compliance:** By integrating QRM with data governance, organizations can proactively address compliance gaps and prepare for inspections and audits.

In the era of Pharma 4.0, where data drives innovation, QRM is essential for ensuring compliance, optimizing operations, and maintaining high product standards. By embedding QRM into data governance, pharmaceutical organizations can harness data's full potential in order to stay competitive while safeguarding patient health.

7.7 Enhancing Data Governance to Meet Evolving Regulatory Requirements

As Pharma 4.0 evolves, robust data governance is essential to meet regulatory expectations from PIC/S [9], EMA [27], FDA [26], and National Competent Authorities (NCAs). With increasing digitalization, regulators emphasize data integrity, accessibility, and traceability. To ensure compliance, pharmaceutical organizations should adopt a structured, future-ready data governance framework built on three key pillars:

- **Performing Data Governance Assessments:** Regular assessments help identify gaps, ensure regulatory alignment, and drive continual improvement. Chapters 6 and 7 explain the key focus areas for this.

- **Establishing a Solid Data Governance Structure:** A well-defined governance framework ensures data accuracy, consistency, and availability throughout the product lifecycle. Essential components for the framework are described in Chapters 6 and 7.

- **Engaging in Open Dialogue with Health Authorities:** Continual engagement with health authorities fosters transparency, keeps organizations informed on regulatory trends, and strengthens compliance. Open communication helps align data governance frameworks with regulatory expectations, reducing risks during audits and inspections.

8 People and Culture

The success of a digital transformation is intrinsically linked to organizational readiness and the supporting cultural framework. Strong and unwavering leadership is essential in setting a strategic direction, prioritizing employee upskilling, promoting cross-functional collaboration, ensuring robust data governance, and nurturing a mindset of continual improvement. These elements form the foundation necessary to address the challenges associated with transformative efforts.

Organizations must embrace new ways of working and thinking. Leadership should cultivate an environment where employees feel empowered to challenge the status quo and to actively work to dismantle functional and departmental silos through open and transparent communication. Additionally, organizations must be candid and transparent in assessing their current digital maturity. Only by acknowledging their present state can they chart a clear path toward achieving their desired end-state. By focusing on these critical areas, organizations can lay the groundwork for a successful digital transformation that drives innovation and growth.

8.1 Organizational Maturity, Culture, and Readiness for Digital Solutions

Ensuring organizational readiness and digital maturity and cultivating a quality mindset and culture for successful Pharma 4.0 implementations involves several best practices, outlined next. By focusing on these practices, organizations can enhance their readiness for Pharma 4.0, improve their digital maturity, and foster a culture that supports innovation and transformation

- **Leadership Commitment:** Strong commitment from top management is crucial. Leaders should actively promote digital transformation initiatives and allocate necessary resources.

- **Communication:** Strategy, vision and clearly articulated goals must be communicated across the organization. Only through clarity, understanding and alignment can a transformation be successful.

- **Empowering Stakeholders:** Stakeholders across all levels must be engaged as key contributors and drivers of sustainable cultural change. As practitioners and process owners, they are well-positioned to implement the vision of their leaders.

- **Clear Vision and Strategy:** Develop a clear vision for Pharma 4.0 and digitalization that aligns with organizational goals. A well-defined strategy helps guide implementation efforts and ensures that everyone understands the objectives.

- **Change Management:** Implement a structured change management process to address resistance and facilitate smooth transitions. Encouraging, promoting, and highlighting early adoption success serves to alleviate concerns and foster buy-in.

- **Process Standardization:** Evaluate existing processes, aligning on a standardized future state before selecting and applying technology. Allowing the process to inform on the technology to be applied ensures alignment with requirements and intended use.

- **Skills Development:** Invest in training and upskilling employees to ensure they possess the necessary digital competencies and skill sets. This can include workshops, online courses, and partnerships with educational institutions.

- **Cross-Functional Collaboration:** Foster collaboration across departments to break down any silos. A multidisciplinary approach can enhance innovation and ensure that different and diverse perspectives are considered in the development and implementation process.

- **Data Governance and Data Management:** Establish robust data governance frameworks to ensure data integrity, security, and accessibility. This is critical for leveraging data analytics effectively in Pharma 4.0. Refer to Chapters 5 and 6 of the EMA Guideline on data management [55] for additional information.

- **Agility and Flexibility:** Encourage a culture of agility, where teams can adapt quickly to changes and experiment with new tools and technologies. This can lead to quicker iterations, better decision-making, and significant process improvements.

- **Customer-Centric Approach:** Focus on understanding and addressing customer needs and pain points. Gathering feedback from stakeholders can help ensure that digital initiatives are aligned with market demands.

- **Proof of Concept:** Start with proof of concept to test new technologies and processes on a smaller scale. Successful proof of concept can help build confidence and provide insights for scaling up to pilot and full production use.

- **Continual Improvement:** Establish mechanisms for continuous monitoring and continual improvement of processes and technologies. Encourage a continual learning mindset, as adaptation is essential for long-term success.

- **Cultural Alignment:** Promote a culture that embraces innovation, collaboration, and transparency. Foster a curious mindset that encourages and supports creativity and trying new things. Encouraging open communication, sharing lessons learned, and recognizing achievements can help reinforce this culture. Allow for the opportunity to create a safe space to “fail fast,” fostering creativity and innovation.

- **Technology Integration:** Establish a clear strategy and plan for the integration of both new and existing technology that aligns with organizational goals. Where practical and feasible, ensure that new technologies integrate seamlessly with existing systems. This can avoid business process disruptions, accelerate technology adoption, and enhance overall efficiency.

- **Historical Expertise:** Leverage the experience and historical knowledge available through long-tenured team members as well as those with related external experiences. Evaluation of diverse perspectives allows for new and different ways of working to be discussed and considered, as well as confirming what cannot or should not change.

8.2 Challenge Pain Points

Realizing the benefits enabled through implementing Pharma 4.0 requires organizations to acknowledge current challenges and commit to their remediation as a priority. A significant digital transformation cannot be accomplished without a focus placed on how people can drive the change necessary to allow teams to unleash the potential offered by new and emerging technologies. Addressing challenges with foundational elements—including a thoughtful and well-conceived people strategy, a continual improvement mindset, and the realization that data is the key that unlocks it all—is core to a successful digital transformation.

Overcoming often long-standing culture, people, and process challenges comes at a price. Conversations with leaders and stakeholders to align prioritizing cleanup and resetting of core foundational principles over implementing the latest technology requires a clear discussion on the impact of not addressing these issues head-on.

- **Centralized System Inventory:** Establish and maintain a consolidated and centralized system/application inventory to ensure a complete awareness of the technology landscape. This inventory should include the interfaces and associations with other systems, ensuring a single source of truth that provides the ability to evaluate and assess risk and impact across end-to-end business processes.

- **Master Data Management:** Ensure the implementation and evolution of a robust MDM program. As the needs of the organization change pertaining to data, the MDM strategy must keep pace to evolve and mature. Cross-functional mechanisms and processes must be appropriately resourced and sustainable to review master data to ensure completeness, accuracy, and availability.

- **Business Relationship Management:** Prioritize the development and growth of strong and sustained relationships with key stakeholders and end users. Changes in processes and the evolving needs of the business must be clearly communicated to ensure the expected benefits are realized through the implementation of new technologies. Exploring a mindset of distributed governance and decentralization (with clear accountability from all involved) can help via a reduced reliance on centralized administration and siloed systems. This enables greater interoperability among business units and encourages stakeholder involvement with new technology implementation.

- **Upskilling the Workforce:** Invest in resource development, as people are the greatest asset. A modernized workforce is critical to the successful implementation of new and emerging technologies. A development roadmap and framework should be implemented to ensure necessary skillsets are available to align with the delivery of technology.

- **Challenging the Status Quo:** Break the mold on legacy thinking. Legacy systems, processes, and ideas may be good, great, and even compliant, but they can often foster a culture of complacency. To take full advantage of what technology can offer, innovation and continual improvement must be core to the new way of working. Evaluating the application of alternative technologies and approaches (e.g., Infrastructure-as-a-Service (IaaS), SaaS, AI/ML) will uncover new possibilities and opportunities that will only be realized through a drive to deliver more.

- **Foster Strategic Thinking:** Promote critical thinking beyond myopic and short-term objectives. Encouraging teams to look beyond the immediate problem to be solved identifies opportunities to be truly transformative. A holistic approach and mindset enables a more thoughtful execution of the strategy and delivery on the roadmap, through scalable and sustainable solutions.

Driving substantive and sustainable cultural change starts with leadership. Although change needs to start at the top of an organization, individual teams and team members can also be leaders and lead by example. Embodying the cultural and behavioral change desired can have a ripple effect, helping to drive change through a groundswell of support from the bottom up.

Embarking on a successful cultural change must start with self-awareness and knowing where every person on every team, and even leadership, sit on the change continuum. Tools and techniques (i.e., surveys, assessments) should be established to enable teams to understand their starting point. This establishes a baseline and allows for a clear path, with milestones, to enable progress (or regression) to be periodically reviewed, discussed, measured, and communicated over time. Only when the current point can be defined and acknowledged, is it possible to chart a course to reach individual, team, and organizational goals.

8.3 Breaking Down Silos

Breaking down organizational silos is essential to foster technology innovation and overcome hindering aspects such as siloed thinking. By doing so, organizations can create a unified, end-to-end approach that aligns with overarching strategies and objectives. This holistic view enables teams to work with a shared purpose, interconnected through cohesive processes. Selecting technology should be based on genuine business needs, addressing well-defined problems rather than chasing the latest trends. A collaborative culture, driven by transparent leadership, facilitates this shift, helping the organization embrace a shared vision. Identifying areas for cross-functional collaboration early ensures successful digital transformation, leveraging organizational strengths and capabilities effectively.

Communicating the “what” and “why” in a way that resonates with the team enables the entire organization to embrace a shared vision. This approach not only enables problem-solving but also cultivates a healthy, collaborative culture that naturally breaks down silos, driving alignment and shared purpose across all levels of the organization.

8.4 Organization Maturity

In Pharma 4.0, achieving organizational maturity is essential for a successful digital transformation. This maturity requires building a people-centric and data-centric organization with a strong cultural foundation.

- The process begins with a clear, end-focused strategy for digital transformation that aligns closely with business needs and incorporates a risk-based management approach.
- By fostering a quality-focused culture, organizations can better position themselves for digital readiness, creating a roadmap that supports continual improvement.
- Establishing this foundation involves robust KM, the identification of organizational pain points, and a future-proof design for long-term resilience, always integrating feedback for ongoing refinement.
- Achieving maturity also requires addressing key prerequisites such as training, organizational alignment, change management, and developing the right mindset to mitigate resistance.
- Maturity should align with business needs, meaning that reaching the highest maturity level is not mandatory for all.

Senior management's commitment is critical, as leadership must drive these efforts to empower their teams and support a digital transformation that enables the organization to adapt, accelerate, and thrive in a Pharma 4.0 landscape.

9 Information Systems

Information systems are the physical and virtual pathways through which data flows, interconnecting the people working in a facility and the equipment within it. The two separate streams of IT and OT come together under Pharma 4.0 to form a heterogenous data flow, connecting the manufacturing floor, laboratory, supply chain, and enterprise systems that live on top of these areas. Through digitalization, this data becomes information and enables the workforce to make more informed decisions to support product quality and patient safety.

In an ideal environment, such as a greenfield site design, it is important to include digital elements early in the conceptual design and planning phases of the project. This ensures that digital elements are included and undergo proper planning themselves—User Requirement Specification (URS), Design Qualification (DQ), etc.—as part of the facility design. This becomes more complex, but not infeasible, in a brownfield environment. Here, additional care and strategic planning are needed to ensure that the digital elements fit seamlessly with those pre-existing elements.

Emerging technologies such as AI, cloud computing, and IIoT can be implemented in compliance with a risk-based approach, such as *ISPE GAMP 5 (Second Edition)* [2]. Additionally, the integration of IT and OT and digital supply chains must align with GxP processes, where *ISPE GAMP 5 (Second Edition)* can serve as a valuable reference for ensuring data integrity, security, and validation/qualification. Guides such as the *ISPE GAMP Good Practice Guide: IT Infrastructure Control and Compliance (Second Edition)* [3], provide strategies and techniques for implementing information systems in a compliant way.

The Pharma 4.0 Operating Model is the conceptual underpinning for all organizational transformation, including information systems. This chapter elaborates on the concepts detailed in the *ISPE Baseline Guide: Pharma 4.0* [1], with particular focus on connectivity and the flow of data throughout the organization and the introduction of emerging technology as a means of increasing digital maturity.

Digital transformation undertaken in a secure, robust, and compliant manner requires an interconnected information web built on a solid infrastructure. This, combined with a collaborative governance framework that supports the drivers of both the IT and the OT groups is essential to support the free and efficient flow of data and the means to convert it to information and knowledge. If this data and information backbone is present, the digitally enabled enterprise can emerge and pharmaceutical manufacturers can reap the benefits and efficiencies that come with it.

9.1 Challenges with Legacy Systems

The challenges with legacy systems outlined in Section 2.6, can be attributed to technical debt. This includes outdated technologies, inflexible architectures, inadequate security, and limited data connectivity. Failure to address these challenges by moving to more current solutions can lead to data silos, complicating data migration, and hindering timely access to information spread across multiple platforms, especially during health authority inspections. Data migration from legacy systems is particularly challenging due to the need to ensure data integrity, accuracy, and completeness. The process often involves transforming data formats, cleaning and validating data, and ensuring compatibility with new systems. Additionally, there is the risk of data loss or corruption during the migration process, which can disrupt operations and lead to compliance issues.

Further challenges with legacy systems include an inability to scale effectively to handle increasing data volumes, evolving business models, or new regulatory compliance demands. The rigid architecture of legacy systems slows down the adoption of modern technologies such as AI, cloud computing, and automation, thereby limiting organizational agility and innovation capacity.

Another concern is that as technology advances, vendors frequently discontinue support, updates, and security patches for legacy systems. This lack of vendor support increases operational risks, exposes organizations to security vulnerabilities, and complicates maintenance. It can also force costly and urgent modernization efforts if critical failures occur, threatening business continuity and compliance.

To address these challenges, a strategic approach is essential, balancing the risks and costs of system upgrades with the long-term benefits of a robust and scalable automation infrastructure. By leveraging new automation technologies, organizations can bridge the gap between legacy systems and modern requirements, ensuring long-term vitality and competitiveness.

Careful strategic planning is also required when migrating from legacy systems, as additional challenges arise when transitioning to modern digital technologies. These challenges include the following:

- **Business Disruption:** Proper business continuity planning is required to keep the plant running while systems are being migrated

- **Data Integrity Concerns:** Migrating complex or data-intensive systems requires forethought to maintain data integrity and prevent data loss

- **Practicality:** Digitalization of paper-based or hybrid solutions can introduce tremendous cost and schedule impacts that may be mitigated by addressing the underlying inefficiencies in the legacy system in advance

With proper advance planning and strategy, these challenges can be mitigated when migrating away from legacy systems.

9.2 Challenges with Data Silos

Data silos, a common phenomenon in legacy systems, present significant challenges in pharmaceutical organizations, particularly when it comes to automation and IT/OT integration. Addressing these challenges requires a strategic approach to data management, including improving data governance, and leveraging modern technologies to ensure seamless data integration and accessibility. Key reasons why data silos present challenges include the following:

- **Fragmented Data:** Data silos lead to fragmented data across different departments and systems. This fragmentation can make it challenging to obtain a comprehensive view of operations, which is essential for effective decision-making and process optimization.

- **Inefficiency and Redundancy:** When data is siloed, different teams may end up duplicating efforts, creating multiple versions of the same data. This redundancy wastes time for resources and increases the risk of errors.

- **Limited Collaboration:** Silos hinder collaboration between departments. For example, R&D, manufacturing, and QC teams may struggle to share critical information, slowing down innovation and problem-solving.

- **Integration Challenges:** Integrating IT and OT systems is essential for automation and real-time data analysis. Data silos complicate this integration, making it harder to implement advanced technologies such as predictive analytics, AI, and IoT.

- **Regulatory Compliance:** Data silos can make it challenging to maintain consistent and accurate records when collating data from multiple disparate systems. These records are necessary for compliance with regulations such as FDA guidelines and GDPR [29].

- **Delayed Time-to-Market:** In a highly competitive industry, speed is crucial. Data silos can delay product development and time to market by slowing down the flow of information and decision-making processes.

- **Internal Compliance:** Silos between IT and OT can create internal policy and procedural challenges due to a lack of process clarity and differentiation. Foundational concepts such as application support and change management are often different, increasing the potential for errors in execution.

- **Delayed Response Time:** Silos can also delay the response to queries from health agencies during inspections, as it can be difficult to find the supporting data. Presenting data from multiple silos can introduce duplication or highlight gaps.

9.3 IT/OT Convergence

The convergence of the IT and OT pillars in a manufacturing organization is necessary to achieve a connected plant engaged in smart manufacturing, which in turn is key to ensuring quality, contextualized data reaches those who need it. In a traditional organization, the IT and OT functions are in separate silos with different business drivers (e.g., uptime and product quality versus information security, data integrity versus reducing costs). Table 9.1 offers a sampling of the differences between traditional IT and OT.

Table 9.1: Differences Between IT and OT [46]

	Information Technology (IT)	Operational Technology (OT)
Focus	Managing data and the flow of information	Controlling physical processes
Priority	Confidentiality	Availability
Lifecycle	3–5 Years	10–20 years
Devices	Off-the-shelf	Purpose built
Data Types	Transactional, structured, and unstructured	Time series, real-time, historical, event, etc.
Hardware	Laptops, tablets, phones, routers, switches, etc.	PLCs, RTUs, HMIs, edge devices, sensors, etc.
Applications	ERP, SCM, BI, HRMS, BPM, CRM, CMS, etc.	PLC, SCADA, MES, DCA, HMI, CMMS, etc.
Standards	ISO/IEC 27000 Series, NIST, CIS, SOC2, COBIT, etc.	ISA-95, ISA/IEC 62443, NIST 800-82, etc.
Workflow Management	NOC/SOC operator workflows	Field operator workflows
Identity Management	User	Device

OT is often the source of the data, the first link in the chain, and the start of the data lifecycle. It is often more efficient to model and contextualize data here, at the "edge", before data is transferred to business systems.

As OT systems become more complex, it becomes harder to distinguish them from IT infrastructure. Distributed SCADA systems, historians, mobile applications for manufacturing, and automated vehicles all rely on IT infrastructure. Functions such as recipe management and batch reporting also straddle the lines between IT and OT. Servers, switches and firewalls, wireless devices, and remote access appliances are commonly found on the shop floor. In the past, this has created a "shadow IT infrastructure" within the OT environment, which leads to a lack of ownership, accountability, and a lack of proper governance. Gaps and overlaps in responsibility are common.

OT has traditionally suffered from a lack of governance, as a myriad of Original Equipment Manufacturers (OEMs) and equipment vendors leads to a lack of standardization and flexibility. Every OEM has its own automation architecture, leverages a specific automation platform, enables a unique means of connecting, and establishes its own data standards. There is often a lack of flexibility from OEMs, which may not follow automation industry standards (e.g., Module Type Package (MTP), ISA-88, ISA-95, ISA-99, ISA-101 [47]). Newer initiatives such as the ISPE Pharma 4.0 Operating Model have yet to be adopted.

As IT organizations have expanded to include new technologies, silos have emerged: network teams handle the physical and logical communications pathways, whereas computer teams are responsible for data centers and clusters of virtual machines. The cloud team enables the enterprise to scale beyond the building's four walls. Cybersecurity makes sure that all systems are safe and secure. Ownership of these various facets of the IT world bleeds together, and when OT is introduced on IT-managed infrastructure, collisions can occur if teams are not working together.

Vendors in both the IT and OT spaces have come together with recommendations to architect networks leveraging best practices and best-in-class solutions. These reference architectures [48] provide guidance to design and deploy a resilient network that considers the criteria and constraints inherent in both the IT and OT worlds.

Updates have been proposed for the Purdue Model of OT and IT to reflect the convergence of IT/OT and to accommodate emerging technologies such as IoT and cloud computing [49].

Cybersecurity and data security come with additional governance requirements. OT systems do not necessarily follow cybersecurity best practices. Unmanaged switches, insecure wireless access points, lack of segmentation, wide open firewalls, the use of legacy software and operating systems, and the reliance on insecure protocols (e.g., DCOM, HTTP) open organizations up to vulnerability. IT may push antivirus onto OT PCs, and this causes systems to fail. The tools that IT uses to manage and secure the network often do not work well when OT systems are involved. A lack of APIs and means of integration forces workarounds and weak points.

Governance at the IT level has become stronger, with several available frameworks available to address various areas, such as the following:

- **Policies and Procedures:** Setting rules that dictate how various tasks and decisions are to be performed
- **Roles and Responsibilities:** Defining the duties and accountability of the various groups and individuals in the organization
- **Decision-Making Processes:** Ensuring consistency and accountability for the decisions made within the organization
- **Risk Management:** Identifying, assessing, and mitigating the various risks that may affect the organization
- **Compliance:** Ensuring adherence to regulatory requirements
- **Performance Monitoring:** Benchmarking and assessing the performance of systems and resources against industry standard indicators
- **Reporting:** Updating stakeholders as to the risk, compliance, and performance of the IT organization

Governance should also be in place on the OT side, with both IT and OT working together to ensure the organization's objectives are met with respect to computerized systems. OT policies, procedures, standards, and reference architectures should be in place to support projects up front, with both IT and OT groups involved throughout the project lifecycles. This drives alignment and buy-in from all groups and ensures success.

Additionally, vendor management is a key area for governance in both IT and OT. *ISPE GAMP 5 (Second Edition)* [2] provides guidance about vendor management for software and computerized system suppliers. The *ISPE GAMP Good Practice Guide: for IT Infrastructure Control and Compliance (Second Edition)* [3] provides guidance for vendor management where IaaS, Platform-as-a-Service (PaaS), or SaaS solutions are deployed. A key area of consideration for vendor management includes the assessment and periodic review of vendors. Activities such as site visits, questionnaires, and audits should be decided upon and leveraged accordingly.

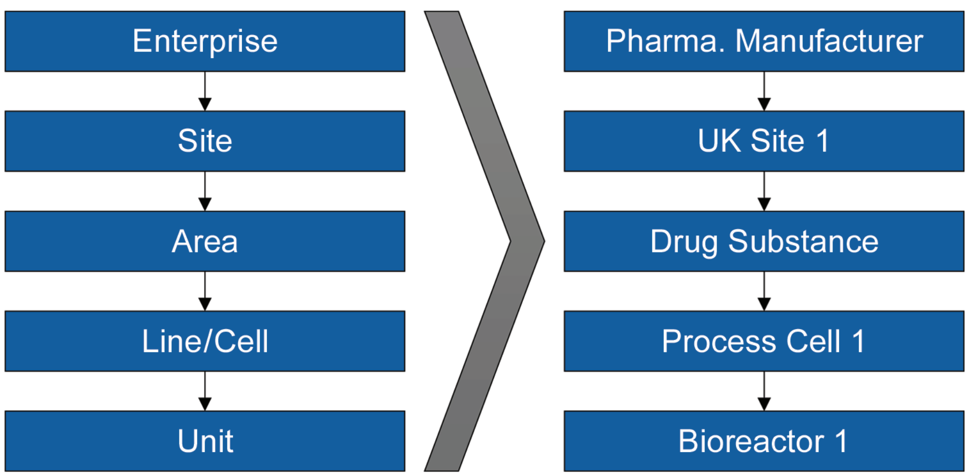
A collaborative approach to governance is recommended. This includes establishing cross-functional teams to combine expertise and better align with a holistic goal of data integrity and continual improvement. A collaborative approach encourages the sharing of best practices and the use of innovative technologies, and it leads to improved regulatory compliance. It ensures the safeguarding of critical systems and data while promoting resilience, quality, and data integrity.

This collaborative approach should extend to tactical activities, such as the response to incidents (e.g., security, downtime). Manufacturing sites should consider establishing a cross-functional critical response team that detects, responds, and reports on incidents pertaining to digital systems and infrastructure, as this will enhance operational effectiveness and minimize the impact of these events.

9.4 Integration Element

One strategy that is becoming more prevalent is leveraging an integrated data management platform. This is a unified data platform—sometimes called a Unified Namespace (UNS) or data hub—for seamless sharing between IT and OT systems to optimize data flow and enhance operational efficiency. This platform enables systems at all levels to communicate by abstracting their interfaces, thus enabling them to speak a common language. It also allows data modeling and contextualization by enabling a semantic hierarchy of data elements that aligns with industry standards such as ISA-95 [18]. An example of contextualized data in a semantic hierarchy is illustrated in Figure 9.1 for a temperature reading from the PLC connected to a bioreactor in a UK biopharma facility.

Figure 9.1: Example of Contextualized Data in a Semantic Hierarchy

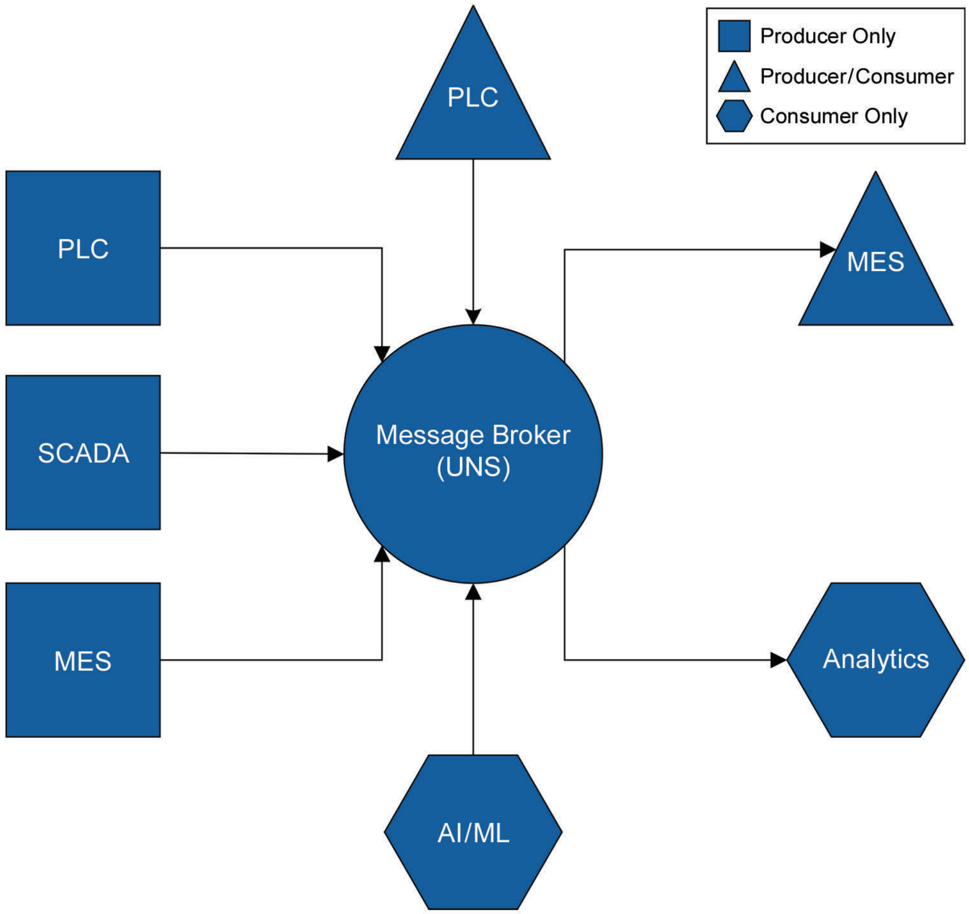


Data Topic: Pharma_Manufacturer/UK_Site_1/Drug_Substance/Process_Cell_1/Bioreactor_1

Data Payload: {"timestamp": 1232343234235
"temperature_degC": 43.2
"connection": "PLC_11"}

Systems publish data to the platform and subscribe to data topics in a hub-and-spoke architecture that reduces point-to-point connections between systems, thus enabling scalability. A detailed description of the UNS is beyond the scope of this Guide, but the concept is illustrated in Figure 9.2. Data producers, consumers, and producer-consumers all integrate with the UNS, the hub of data architecture.

Figure 9.2: The Concept of a Unified Data Platform



9.5 The Connected Plant and Smart Manufacturing

To ensure good plant connectivity and operational effectiveness in a smart manufacturing facility using Pharma 4.0 technologies and capabilities, consider applying the following best practices related to information systems:

- **Robust Network Infrastructure:** Invest in a reliable, redundant, and high-speed network infrastructure to support data transfer between devices, systems, and cloud services. This includes both wired and wireless solutions to ensure comprehensive coverage.

- **Interoperability Standards:** Adopt industry standards and frameworks (such as ISA-95 [18] and the UNS) to ensure that different systems and devices can communicate effectively. This facilitates seamless integration across various platforms and systems.

- **Real-Time Data Acquisition:** Implement systems that allow for real-time data collection and monitoring from various types of equipment and processes. This enables timely decision-making and enhances operational visibility.

- **Cloud Integration:** Utilize cloud-based solutions for data storage and processing. Cloud platforms can offer scalability, flexibility, and access to advanced analytics and ML capabilities.

- **IoT Device Management:** Deploy IoT devices and sensors for enhanced visibility and control. Ensure that there are robust management protocols for maintaining and securing these devices.

- **Data Analytics and Visualization Tools:** Incorporate advanced analytics and visualization tools to transform raw data into actionable insights. This can include dashboards that provide real-time insights into operations and key performance metrics.

- **Digital Twins:** Leverage virtual models of physical assets and processes to enable simulation, scenario planning, and continual improvement. Digital twins allow manufacturers to test proposed changes, predict equipment failures, and optimize production without disrupting the manufacturing operation. This drives efficiency through a reduction in downtime and supports data-driven decision-making.

- **Cybersecurity Measures:** Implement strong cybersecurity protocols, including those established under the ISA/IEC 62443 series of standards [35], to protect sensitive data and ensure the integrity of manufacturing systems. Regularly update software and conduct vulnerability and risk assessments. Leverage threat modeling in the design phase to reduce attack surfaces and eliminate single points of failure.

- **Centralized Data Management:** Establish a centralized data management system to streamline data access and governance. This helps ensure consistency and reliability of data across the entire organization.

- **Flexible Manufacturing Systems:** Invest in flexible and modular systems that can be easily adapted to changing production needs. This allows for quick adjustments in response to market demands, regulatory changes, and other external factors.

- **User Training and Support:** Provide comprehensive training for employees in any new information systems and technologies. Continual support and resources can help users maximize the benefits of these new systems.

- **Collaboration Tools:** Use collaboration tools that facilitate communication between different teams and departments. These tools can help ensure that all stakeholders are aligned and informed about all aspects of the operations.

- **Continuous Monitoring and Feedback:** Implement systems for continuous monitoring of plant performance and user feedback. This can help identify areas for improvement and drive ongoing optimization and improvement efforts.

- **Regulatory Compliance:** Ensure that all information systems comply with industry regulations (such as FDA [26], EMA [27], and other regulations in the pharmaceutical industry). This includes maintaining accurate records and ensuring traceability.

By adhering to these best practices, organizations can enhance plant connectivity and operational performance, ultimately resulting in successful implementations of Pharma 4.0s initiatives within smart manufacturing environments.

There are many benefits to implementing a connected plant, which involves the integration of business and manufacturing systems (often called “shop floor to top floor”). BioPhorum provides a Digital Plant Maturity Model that categorizes the connected plant as maturity level 3 within the following levels [50]:

- Maturity level 1 – Pre-digital plant – Manual, paper-based processes

- Maturity level 2 – Digital islands – ‘Islands of automation’

- Maturity level 3 – Connected plant – High-level of automation, integration and systems standardization

- Maturity level 4 – Predictive plant – Integrated plant network, pervasive real-time predictive analytics

    - *“Represents the highest level of automation and integration achievable with current technology”*

- Maturity level 5 – Adaptive plant – ‘Plant of the future’, autonomous, self-optimizing, plug-and-play

Some of the benefits of the connected plant include:

- **Reduced Manual Effort:** In a traditional plant, manufacturing data resides in shop floor systems or, at best, databases and historians. It is in a company's business systems, where sales orders are received and production planning takes place. Manual effort is required to extract manufacturing data so that production planning and scheduling personnel can understand the plant's capacity. Similarly, converting planned production into actual production requires manual order retrieval and data entry to the manufacturing systems. Connecting the two (typically via a MES system) creates efficiency through the reduction of manual effort in the data flow process.

- **Increased Traceability:** Within the walls of the manufacturing site, connectivity between plant floor and enterprise systems enables data to flow between these systems. This enables better traceability from when raw materials and components enter the building to when finished goods leave and increases the robustness of the QMS. Extending this beyond the facility, connectivity to other entities creates a value network that includes suppliers, manufacturers, supply chain partners, regulators, and customers.

9.6 Underlying Infrastructure Requirements

When considering the infrastructure required to support digital systems, it is important to think of the entire data lifecycle and data integrity by design. Information systems need to live in a connected web that includes the data producers and consumers (sources and sinks). This includes operators, laboratory personnel, maintenance, warehouse personnel, data scientists, analysts, QA, and others. There is a plethora of data sources in the connected plant, ranging from individual sensors to laboratory instrumentation to complex manufacturing equipment to monolithic business software. The infrastructure needs to enable the data to flow in a secure and robust way.

Infrastructure design is use-case specific, but should be aligned to the common objective of secure and robust data flow across multiple systems to be shared with multiple stakeholders. Various approaches address this, such as data hubs and self-sovereign distributed architectures. The decision of whether to take a centralized approach or to explore decentralized strategies lies with the organization. That being said, manufacturing data should be saturated with enough business and process context to enable all stakeholders to have a complete picture of the organization and so that the applications and interfaces leveraging the data are able to create tangible value for the users.

The infrastructure underpins the production and consumption of data, but the applications and interfaces are decoupled and abstracted from it. This allows the users to work directly with fully contextualized and modeled data so that, for example, a data analyst does not need to understand the technical details of the OT paradigm (e.g., PLC addresses, OPC tags) to be able to use it. This infrastructure is similar to a building's plumbing: it is in the walls of the building, and provides functionality to the residents inside, though they do not interact with it.

To achieve this, the infrastructure must be secure, scalable, and have open interfaces to a variety of software and systems. It should be simple to maintain and manage, including both the software and hardware elements. Additionally, the infrastructure should be qualified and operated in a controlled manner. The impact of any changes must be assessed to avoid unexpected downtime due to unintended consequences of those changes. The more interconnected the facility becomes, the more people rely on digital systems to perform routine tasks. Planning to avoid downtime and loss of connectivity is key to the digital factory.

10 Practical Use Cases

This chapter provides a comprehensive overview of the implementation of various Pharma 4.0 enabling technologies. It delves into the specific technologies and solutions, highlighting their benefits, challenges, and associated facility infrastructure requirements. Identifying these infrastructure requirements is crucial, as they significantly influence the design of a facility. The design considerations span multiple disciplines, including but not limited to the following (with examples):

- Architecture Layouts: Additional space requirements for charging stations, wider hallways, wall space for displays, etc.
- Door Controls: An Automated Guided Vehicles (AGV) requires integration with the door controls so that they open for the AGV
- Electrical Design: Additional power outlets may be required
- Automation Design: More data points may be required
- IT: A more robust network architecture design may be required

Understanding the impact of these factors is essential to successfully integrate Pharma 4.0 technologies into modern pharmaceutical facilities. These use cases are meant to offer guidance on items to consider before implementing technologies. Importantly, if the problem statement does not resonate or is not a pain point in the facility, it may not have the desired outcome. Conversely, if the problem statement resonates, it may be something to investigate further. The benefits and challenges should be quantified for the unique use case(s) being addressed at a specific facility.

These use cases are not intended to be an exhaustive list. They present ideas on technologies that are being implemented at scale by manufacturers and are meant to highlight the considerations of each technology across the Pharma 4.0 Operating Model and identify potential benefits and challenges. Technologies should be addressed and analyzed as part of a comprehensive digital roadmap that is supported by a digital maturity assessment, which is updated as progress is made. This will ensure that technologies are not adopted in search of a problem, but that they support a clear business objective that has been aligned with stakeholders.

Particular care should be taken in enterprise-wide rollouts that need to balance scale and innovation. A corporate-wide Pharma 4.0 strategy must include the ability to scale. It is important to note the importance of strategic positioning and the competitive differentiation associated with Pharma 4.0, emphasizing that digital transformation is not just about compliance, but rather about building long-term value. By understanding the impacts and requirements of digital maturity for Pharma 4.0 enabling technologies, organizations can understand the prerequisites and infrastructure necessary and ensure their strategy does not incur technical debt.

10.1 Integrating Autonomous Vehicles

Problem Statement

Safety, efficiency, accuracy, and reliability concerns are inherent in human-driven inter- and intra-building material transport and delivery. Operators and logistics resources rarely transport goods using the most efficient path, or even a consistent path. Damage to material, equipment, and the building itself is often caused by personnel moving heavy loads. Injuries are likely to happen and ergonomic issues arise over time. Manual movement of materials necessitates additional manual data entries (i.e., logbooks or manual entries into WMSs). Further, in aseptic areas, people always pose a contamination risk.

Solution

Automated conveyance, autonomous vehicles, and Automated Storage and Retrieval Systems (ASRS) can be used to reduce human error, increase safety, and increase efficiency in material movement. Autonomous vehicles include Automated Guided Vehicles (AGVs) and Autonomous Mobile Robots (AMRs).

ASRSs are autonomous vehicles used in conjunction with warehouse racking to automate the storage and retrieval of goods in a warehouse. They typically work in conjunction with conveyors to enable pallets to be picked up and dropped at one or more static locations. ASRSs enable lights-out warehousing, with little-to-no human intervention.

AGVs are typically used when transport is via pallet jack or fork truck. They usually have fixed preprogrammed routes and minimal onboard intelligence. They connect back to a central orchestration system to receive their next job or task. Onboard safety systems cause them to slow down and eventually stop in the proximity of any object (e.g., person, structure, equipment); they will restart when the object or person is removed from their movement envelope.

AMRs are typically used when the load is picked up from underneath. They can work autonomously from the central orchestration system and navigate with pre-programmed maps. Flexible routing allows AMRs to maneuver around obstacles. They are capable of lower-weight payloads (up to about 1,000 kg). Further, Mobile Manipulator (MoMa) robots consisting of a six-axis cobot on top of an AMR can be used to perform many tasks, such as loading and unloading in aseptic areas or isolators.

Benefits

- Increased efficiency and accuracy in material handling – right-first-time delivery
- Reduction in logistics-based labor cost – fewer warehouse and logistics technicians are required
- Enhanced safety in the transportation of materials – automated vehicles have sensor networks to prevent crashes and damage to both people and facilities
- Reduced footprint for material storage when ASRSs are used
- Reduced contamination risk in aseptic environments
- Improved traceability with automatic versus manual movement of materials

Challenges

- High initial investment cost – Return on Investments (ROI) depends on throughput and scale
- Integration with existing systems – autonomous vehicles derive the most value when they are integrated with WMS and ERP systems
- Facility design considerations – floor surface and grade must be appropriate for vehicles; corridors and rooms must be sized to allow vehicles to move around
- Robot cleanability – most robots are not designed for pharmaceutical environments and have not been evaluated with decontamination equipment; wipedowns can also be time consuming
- New integration protocols or software layers – Using autonomous vehicles requires new software systems to manage them and potentially new communication protocols to accomplish this (i.e., MQTT)

Facility Design Factors to Consider

- Wireless communications and navigational infrastructure to support autonomous vehicles

- Charging stations for autonomous vehicles
- Safety protocols for human-vehicle interaction
- Transition of robots or materials between different room classifications

Impact On

- **Data:** Increased data collection on logistics and transport efficiency
- **People and Culture:** Need for training staff to maintain and work alongside autonomous systems
- **Informational Systems:** Wireless communications and integration with warehouse management and logistics systems
- **Regulatory:** Compliance with safety and operational regulations for autonomous vehicles

10.2 Location-Based Technology

Problem Statement

There is difficulty in tracking raw materials, Work-in-Progress (WIP) materials, and assets, and difficulty ensuring cold chain integrity.

Solution

Real-time Location Systems (RTLS) enable tracking of materials and assets, and geofencing enables monitoring of abnormal asset movement. This can be particularly beneficial for portable, high-value assets.

Benefits

- Improved asset tracking and management
- Enhanced visibility into raw material, WIP, and finished goods inventory
- Reduced risk of asset misplacement

Challenges

- Cost of deployment of RTLSs, involving significant investment in hardware, software, and infrastructure
- Technical limitations such as signal interference, limited coverage, and accuracy issues; these systems rely on wireless technologies, and each has benefits and drawbacks, depending on the application
- Integration with existing inventory systems can be a challenge due to the issues inherent with information system integration; careful planning and vendor openness are required to ensure success

Facility Design Factors to Consider

- Placement of sensors and wireless infrastructure; a wireless coverage mapping exercise is necessary
- Training and maintenance, with ongoing support for the RTLS; specialized training to ensure staff are capable of using the system effectively

- Easy access to sensors and components

Impact On

- **Data:** Real-time data on asset location and condition
- **People and Culture:** Need for training on new systems and processes
- **Informational Systems:** Integration with inventory and supply chain management systems
- **Regulatory:** Ensuring compliance with regulations for asset tracking and cold chain management

10.3 Artificial Intelligence Use Cases

AI is a rapidly changing and emerging topic, and several use cases are summarized next. For additional industry guidance, refer to the *ISPE GAMP® Guide: Artificial Intelligence* [51]. For regulatory aspects, EudraLex Annex 22 (draft) addresses Artificial Intelligence [52]. This list includes various examples of how AI is being used in industry but is not exhaustive.

10.3.1 Operations Supported by AI

Problem Statement

There are inefficiencies in drug discovery, clinical trials, and supply chain management. Quality management is difficult, manual, and time consuming (both with corrective actions and identifying trends).

Solution

Implement AI for predictive analytics, patient recruitment, supply chain, and QMS optimization. Drug discovery has received the most attention for AI use cases, helping to bring medicines to clinical trial faster. Many manufacturers are using AI to better understand manufacturing assets. Similarly, using quality management data to perform analysis can provide unexpected insights and greatly reduce the cost of generating documentation related to quality events.

Benefits

- Faster drug discovery and development
- Improved patient recruitment and trial efficiency
- Optimized inventory and logistics
- Reduced time spent resolving quality incidents and trends

Challenges

- Need to address data governance and strategy before scaling AI systems
- Data privacy and security concerns
- High implementation costs
- Need for skilled personnel to manage AI systems

Facility Design Factors to Consider

- Data infrastructure to support AI analytics
- Secure storage for sensitive data
- Integration with existing research and operational systems

Impact On

- **Data:** Enhanced data analytics capabilities
- **People and Culture:** Need for AI literacy and training – organizational units need to be developed to support and drive AI use cases; agile methodology should be integrated into project execution
- **Informational Systems:** Integration with research and operational systems
- **Regulatory:** Compliance with data privacy and security regulations; operating procedures need to be updated with clear guidance when AI is performing a GMP function

10.3.2 Smart Scheduling

Problem Statement

Scheduling systems are disconnected from the factory floor, leading to batches falling behind schedule and uncertainty within supply chains (both upstream and downstream). This is especially challenging with personalized medicines, with one batch per one patient.

Solution

Smart scheduling systems use AI/ML or other algorithms to understand the current state of the factory floor and make predictions on the future state. They can expedite batches in progress, understand when production is bottlenecked, and better align upstream and downstream stakeholders with real-time schedule updates.

Benefits

- Understand production schedules in real-time
- Reduced batch cycle time through better visualization
- Better communication with warehouse functions
- Better understanding of human resource requirements
- Understand cause and effect of equipment downtime
- Better integration with supply chain (raw and intermediate materials)

Challenges

- Requires complicated integration with MES (or similar) systems
- Time-to-value can be slow as the system collects data (though this can be mitigated if the AI algorithm can be trained with good data in advance)

Facility Design Factors to Consider

- Data infrastructure to support integration with MES

Impact On

- **Data:** Data generated from MES or batch record systems is critical and can be used for training if legacy data exists
- **People and Culture**: Scheduling personnel change from a reactive to proactive approach; coordination between manufacturing, warehouse and scheduling teams is important
- **Informational Systems:** Detailed integration plans with MES need to be considered at project outset to ensure the correct data is available
- **Regulatory:** Scheduling is generally not a GMP function, so regulatory requirements are minimal

10.3.3 Document Generation

Problem Statement

Pharmaceutical manufacturing requires robust documentation to support operating in a regulated environment. Generating this documentation is labor intensive and expensive.

Solution

By using LLMs trained on good documentation and supported with other digital design tools (such as BIM), documents can be substantially generated with AI, with humans used to review, refine, and improve. Examples include (but are not limited to) URS, commissioning and validation protocols, vendor GAMP documents, and SOPs.

Benefits

- Reduced cost and time associated with document generation
- More consistent documents generated based on best-in-class examples with lessons learned
- Humans used to review and improve content of documents instead of on time-consuming, repetitive tasks (such as formatting)

Challenges

- Good datasets are hard to find and are generally proprietary; even within a single company many different standards are used which may limit the effectiveness of the LLM and cause additional work in human review cycles

Facility Design Factors to Consider

None

Impact On

- **Data:** Document set for training must be carefully curated

- **People and Culture:** Changing mentality from “doer” to “reviewer” and learning to trust and question AI models and their output
- **Informational Systems:** Detailed data governance and infrastructure to support document generation
- **Regulatory:** Since document review and approval are still human functions, no additional regulatory hurdles are anticipated

10.3.4 QA System Trends, Reporting, and Analysis

Problem Statement

Quality functions are critical but often rely on labor-intensive analysis and individual SMEs to perform analysis and identify systemic issues and root causes. Further, information is often siloed in different organizational units and facilities and is not shared across the enterprise.

Solution

Train AI models on QMSs – including quality incidents, laboratory incidents, Corrective Actions and Preventive Actions (CAPAs), alongside KM systems and data used to support investigations. AI tools can be built to monitor trends in incidents and to share best practices and lessons learned across an organization This can help guide teams toward root cause analysis based on similar incidents.

Benefits

- Much better understanding of quality trends across an area, site, and organization
- Real-time updates on incidents can be shared easily and correlated instead of in quarterly reports
- More consistent quality outcomes
- Reduced time in documentation generation

Challenges

- Datasets are limited and correlating data to achieve the desired outcomes is very difficult

Facility Design Factors to Consider

None

Impact On

- **Data:** Document set for training must be carefully curated
- **People and Culture:** Changing mentality from “doer” to “reviewer” and learning to trust and question AI models and their output
- **Informational Systems:** Detailed data governance and infrastructure to support document generation
- **Regulatory:** Any corrective actions or incident responses need to be carefully validated alongside a risk assessment

10.4 Use of Augmented and Virtual Reality for Training

Problem Statement

There is a need for effective training methods for complex processes and easy access to real-time guidance while performing operations. Subject matter expertise may be spread globally, and companywide expertise can be lost at a plant level.

Solution

By implementing Virtual Reality (VR), manufacturers can enhance their training programs to be more effective. Training can be done in an office environment instead of a cleanroom, standardized across the company, and developed by centers of excellence. Similarly, by using Augmented Reality (AR), operators can be guided in real time by remote SMEs or have their training materials digitally retrieved while performing a task. Further, by implementing VR training early in the process (such as during construction of a facility or when equipment is being designed), operators can be ready to support production on day one and design issues identified earlier.

Benefits

- Enhanced learning and retention
- Safe environment for practicing complex procedures
- Reduced training costs over time
- Improves operational readiness
- Attracts talent

Challenges

- High initial setup costs
- Need for specialized equipment and software
- Ensuring realistic and accurate simulations
- Certain health conditions, such as epilepsy can restrict users’ ability to safety use VR – companies must consider this as part of their training plan

Facility Design Factors to Consider

- Space for VR training setups (i.e., VR training room)
- Network infrastructure to support AR/VR applications
- Safety measures for VR training environments

Impact On

- **Data:** Collection of training performance data
- **People and Culture:** Increased engagement and skill development; human factors should also be considered in developing the training environment since some people are sensitive to VR

- **Informational Systems:** Integration with training management systems
- **Regulatory:** Ensuring training programs meet regulatory standards

10.5 Digital Control Towers

Problem Statement

There is a lack of real-time visibility and control over supply chain and production processes that are end to end.

Solution

Implement digital control towers for real-time monitoring and management. A digital control tower uses a connected data infrastructure (i.e., a UNS) to leverage data from multiple sources in the plant (e.g., manufacturing, QC, warehouse, Computerized Maintenance Management System (CMMS), ERP) and combine it to present a holistic view of the enterprise via dashboards and other visualization tools. It can help personnel understand what is happening in the plant (and when extended to external partners, across the supply chain, for example) in real time.

Benefits

- Provides visibility to the overall process
- Improved supply chain efficiency and responsiveness
- Enhanced QC and risk management
- Better decision-making with real-time data
- Helps with planning and management of deviations and events

Challenges

- High implementation and maintenance costs
- Integration with existing systems
- Ensuring data accuracy and reliability
- Extension beyond the four walls of the factory requires collaboration with external partners

Facility Design Factors to Consider

- Centralized location to mount the displays
- Network infrastructure for real-time data transmission
- Security measures for data protection

Impact On

- Data: Real-time data collection and analysis
- People and Culture: Need for training on new systems and processes

- Informational Systems: Integration with supply chain and production systems
- Regulatory: Compliance with data security and operational regulations

10.6 Digital Signage and Labels

Problem Statement

Dynamic and up-to-date information display for equipment, materials, and rooms are needed in facilities.

Solution

Use digital signage and labels for real-time information updates. These systems remove the need for paper signs and labels, leveraging digital display and e-ink technology. Similar to the digital control tower, these systems use a connected data infrastructure (i.e., a UNS) to leverage data from multiple sources for display purposes. Figure 10.1 shows an example of a digital sign and a digital label.

Figure 10.1: Digital Signage and Label
Images courtesy of Systec & Solutions GmbH, www.systec-solutions.com.

Benefits

- Easy to make updates and changes to information
- Improved compliance with regulatory requirements
- Enhanced personnel communication

Challenges

- Initial setup and maintenance costs
- Ensuring accuracy and timeliness of displayed information
- Integration with existing information systems

Facility Design Factors to Consider

- Placement of digital signage for optimal visibility
- Network infrastructure to support real-time updates

- Security measures for information integrity

Impact On

- **Data:** Real-time information display and updates
- **People and Culture:** Improved communication and information access
- **Informational Systems:** Integration with enterprise and manufacturing systems
- **Regulatory:** Ensuring compliance with information display regulations

10.7 Operator Interfaces

Problem Statement

Efficient and accurate process control is needed in manufacturing and QA. Information is often kept in silos, preventing fast decision making.

Solution

Implement modern operator interfaces for real-time data and control. Operational interfaces should allow people to feel empowered to make good decisions quickly. By designing interfaces that break down data silos and provide important context, interfaces can also support continual improvement and root cause analysis.

Benefits

- Improved process control and efficiency
- Enhanced accuracy in QA
- Reduced human error
- Empowered people

Challenges

- High implementation and training costs
- Integration with existing systems
- Ensuring user-friendly interface design
- Design of interfaces such that they accurately display insights instead of noise.

Facility Design Factors to Consider

- Ergonomic design of operator workstations.
- Network infrastructure for real-time data access.
- Security measures for system integrity.

Impact On

- **Data:** Real-time process and quality data.
- **People and Culture:** Need for training on new interfaces.
- **Informational Systems:** Integration with manufacturing and quality systems.
- **Regulatory:** Compliance with process control and QA regulations.

10.8 Biometrics

Problem Statement

Secure and accurate personnel identification and access control are needed. Specifically, eBRs can be very cumbersome and require operators and supervisors to enter their username and password hundreds of times per day.

Solution

Use biometric authentication for personnel identification and facility access. Biometrics are physical characteristics that are unique to each individual. One example of biometrics is identification bands worn around the wrist (such as a watch) that are digitally linked to the wearer via fingerprint and heartbeat. Another example is iris-based visual recognition, which recognizes the unique ring-shaped area of the eye around the pupil when users look into a camera for identification. These systems enable password-less and token-less means of authenticating a user's identity both for logging onto a system or for electronic signatures.

Privacy of PII and other aspects covered by laws such as GDPR [29] should also be a priority when implementing biometric solutions. Biometric solutions can serve as electronic signatures compliant with 21 CFR Part 11 [11] and other electronic signature regulations, reducing the operational frustration from multiple password entries.

Benefits

- Enhanced security and accuracy in patient identification
- Reduced risk of fraud and errors
- Improved access control

Challenges

- Privacy and data security concerns
- High implementation costs
- Ensuring accuracy and reliability of biometric systems
- Awareness of cultural aspects to ensure user adoption
- Development of risk assessment and response plans for managing PII

Facility Design Factors to Consider

- Placement of biometric scanners and sensors

- Network infrastructure for data transmission
- Security measures for data protection

Impact On

- **Data:** Secure and accurate identification data
- **People and Culture:** Need for training in biometric systems
- **Informational Systems:** Integration and interoperability with information systems (e.g., access management, active directory, OT applications)
- **Regulatory:** Compliance with data privacy and security regulations

10.9 Digital Twins

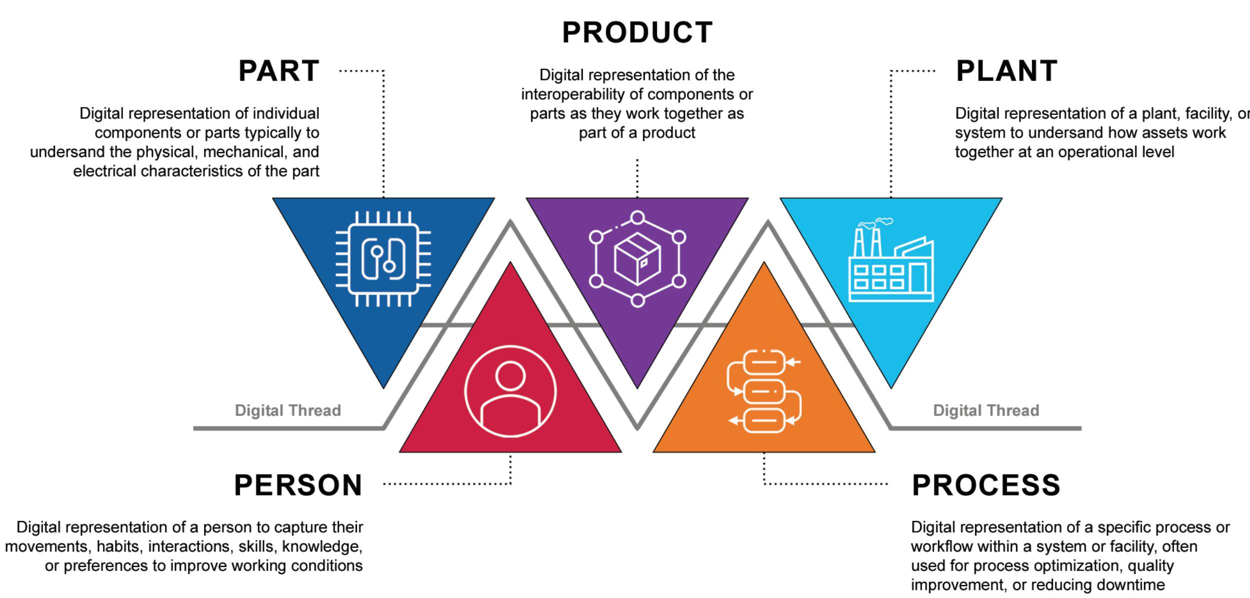
Problem Statement

Designing manufacturing facilities and optimizing manufacturing processes, operator training, construction, and drug development can be time-consuming, costly, and inefficient.

Solution

Develop in silico models of components, products, manufacturing facilities, operators, and processes to enable simulation and drive efficiency. These models are digital representations, also known as digital twins, as shown in Figure 10.2. For example, a digital twin of an aseptic filling line enables an operator to virtually interact with the line, particularly when used in conjunction with AR/VR, to enable training without occupying time on the actual machine and creating a loss of sterility within the machine envelope. Coordinating with BIM modeling can detect clashes early, reducing costs and developing more detailed and accurate schedules. Standards for interoperability of digital twins, such as ISO 23247 [53], should be considered as part of a digital twin strategy.

Figure 10.2: Five Types of Digital Twins [54]
Used with permission, www.jeffwinterinsights.com.



Benefits

- Faster drug development and reduced costs
- Improved manufacturing efficiency and quality
- Enhanced clinical trial design and simulation
- Improved construction and commissioning/qualification/validation activities when linking physical and digital models

Challenges

- High implementation and data management costs
- Integration with existing systems
- Ensuring accuracy and reliability of simulations

Facility Design Factors to Consider

- Data infrastructure to support digital twin simulations
- Secure storage for simulation data
- Integration with research and operational systems

Impact On

- **Data:** Enhanced simulation and optimization data
- **People and Culture:** Need for training on digital twin technologies
- **Informational Systems:** Integration with research and manufacturing systems
- **Regulatory:** Compliance with data security and simulation accuracy regulations

10.10 Predictive Maintenance

Problem Statement

Unexpected equipment failures or performance degradation cause unplanned downtime and extra maintenance costs.

Solution

Use predictive maintenance to monitor equipment reliability and lifespan. Many use cases for predictive maintenance are well-understood, especially for rotating equipment. Monitoring a motor for temperature and vibration has well-understood failure modes and high accuracy, and many sensors and third-party applications that support this use case. It is one of the most well-defined ML use cases. However, for more complicated equipment, failure modes are more difficult to predict, require custom instrumentation, and do not have commercially available third-party platforms.

Benefits

- Enhanced operational efficiency
- Reduced downtime
- Ability to anticipate problems before they occur, and then set up maintenance to resolve the issue before it becomes a problem

Challenges

- Requires designing a gateway network for IoT-based systems or wiring new instruments to existing control systems
- Integration with existing maintenance systems
- Ensuring accuracy of predictive models
- Must use a third-party platform or develop customized analytics
- Modeling complex equipment

Facility Design Factors to Consider

- Placement of gateways for optimal monitoring; generally, sensors are wireless and connect to gateways to send data to an analytics platform
- Network infrastructure for real-time data transmission
- Security measures for data protection

Impact On

- **Data:** Real-time equipment health data
- **People and Culture:** Need for training in predictive maintenance systems and a deep understanding of complex failure modes
- **Informational Systems:** Integration with maintenance management systems
- **Regulatory:** Compliance with maintenance and safety regulations

10.11 360-Degree Cameras and Laser Scanners

Problem Statement

Accurate, efficient, and comprehensive documentation and analysis is needed for of existing facilities that are being renovated as well as construction sites and facilities.

Solution

The suggested solution is to implement 360-degree cameras and laser scanning technologies to capture detailed and panoramic views of environments, enabling precise documentation, analysis and management.

Benefits

- 360-Degree Cameras:
  - Fast time capture
  - Cost-effective and quick to perform
  - Provide construction progress documentation
- Laser Scanners:
  - High precision
  - Integration with BIM

Challenges

- 360-Degree Cameras:
  - Resolution is slightly less than laser scanning
- Laser Scanning:
  - High initial investment
  - Specialized training and expertise required
  - Integration with existing systems can be complex

Facility Design Factors to Consider

- Data Infrastructure: Robust infrastructure is needed to support large data sets from 360-degree cameras and laser scanners
- Secure Storage: Secure and scalable storage solutions for the captured data
- Integration: Seamless integration with existing operational and BIM systems
- Accessibility: Ensuring data is available and easily accessible to relevant stakeholders

Impact On

- **Data:** Enhanced accuracy and comprehensiveness of simulation and modeling data
- **People and Culture:** Need for training in new technologies and fostering a culture of innovation
- **Informational Systems:** Integration with design, construction, and facility management to streamline workflows
- **Regulatory:** Compliance with data security regulations and ensuring the accuracy and reliability of captured data and simulations

11 Appendix 1 – References

1. *ISPE Baseline® Pharmaceutical Engineering Guide, Volume 8 – Pharma 4.0™,* International Society for Pharmaceutical Engineering (ISPE), First Edition, December 2023, www.ispe.org.

2. *ISPE GAMP® 5: A Risk-Based Approach to Compliant GxP Computerized Systems,* International Society for Pharmaceutical Engineering (ISPE), Second Edition, July 2022, www.ispe.org.

3. *ISPE GAMP® Good Practice Guide: IT Infrastructure Control and Compliance,* International Society for Pharmaceutical Engineering (ISPE), Second Edition, August 2017, www.ispe.org.

4. *ISPE Guide Series: Advancing Pharmaceutical Quality, Cultural Excellence,* International Society for Pharmaceutical Engineering (ISPE), First Edition, November 2022, www.ispe.org.

5. US Food and Drug Administration (FDA), "Emerging Technology Program (ETP)," 24 February 2025, www.fda.gov/about-fda/center-drug-evaluation-and-research-cder/emerging-technology-program-etp.

6. European Medicines Agency (EMA), "Innovation Task Force briefing meetings," Accessed 8 October 2025, www.ema.europa.eu/en/human-regulatory-overview/research-development/innovation-task-force-briefing-meetings.

7. European Medicines Agency (EMA), "Quality Innovation Group," Accessed 8 October 2025, www.ema.europa.eu/en/committees/working-parties-other-groups/chmp-working-parties-other-groups/quality-innovation-group.

8. ICH Harmonised Guideline: *Pharmaceutical Quality System – Q10,* Step 4, 4 June 2008, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.

9. Pharmaceutical Inspection Co-operation Scheme (PIC/S), www.picscheme.org.

10. PIC/S Guidance: Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments, 1 July 2021, Pharmaceutical Inspection Co-operation Scheme (PIC/S), www.picscheme.org.

11. 21 CFR Part 11 – Electronic Records; Electronic Signatures, Code of Federal Regulations, US Food and Drug Administration (FDA), www.fda.gov.

12. EudraLex Volume 4 – Guidelines for Good Manufacturing Practices for Medicinal Products for Human and Veterinary Use, Annex 11: Computerised Systems, Manufacture of Sterile Medicinal Products, January 2011, https://health.ec.europa.eu/medicinal-products/eudralex/eudralex-volume-4_en.

13. EudraLex Volume 4 – Guidelines for Good Manufacturing Practices for Medicinal Products for Human and Veterinary Use, Annex 15: Qualification and Validation, 30 March 2015, https://health.ec.europa.eu/medicinalproducts/eudralex/eudralex-volume-4_en.

14. *ISPE GAMP® Guide: Records and Data Integrity,* International Society for Pharmaceutical Engineering (ISPE), First Edition, March 2017, www.ispe.org.

15. ICH Harmonised Guideline: *Quality Risk Management – Q9(R1),* Step 4, 18 January 2023, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.

16. Minero, T. and Kuger, L., "The 8th ISPE Pharma 4.0™ Study," 2025 ISPE Europe Annual Conference, 12–14 May 2025, London, United Kingdom, www.ispe.org.

17. *ISPE Good Practice Guide: Knowledge Management in the Pharmaceutical Industry,* International Society for Pharmaceutical Engineering (ISPE), First Edition, May 2021, www.ispe.org.

18. ISA-95 Series of Standards, International Society of Automation (ISA), www.isa.org/standards-and-publications/isa-standards/isa-95-standard.

19. OPC® Foundation (Open Platform Communications), www.opcfoundation.org.

20. PIC/S Guide to Good Manufacturing Practice for Medicinal Products, Annex 11 Computerised Systems, 25 August 2023, Pharmaceutical Inspection Co-operation Scheme (PIC/S), www.picscheme.org.

21. GO FAIR International Support & Coordination Office, "FAIR Principles," Accessed 8 October 2025, www.go-fair.org/fair-principles/.

22. Regulation (EU) 2024/2847 of the European Parliament and of the Council of 23 October 2024 on horizontal cybersecurity requirements for products with digital elements and amending Regulations (EU) No. 168/2013 and (EU) 2019/1020 and Directive (EU) 2020/1828 (Cyber Resilience Act), 20 November 2024, https://eur-lex.europa.eu/eli/reg/2024/2847/oj/eng.

23. Directive (EU) 2022/2555 of the European Parliament and of the Council of 14 December 2022 on measures for a high common level of cybersecurity across the Union, amending Regulation (EU) No 910/2014 and Directive (EU) 2018/1972, and repealing Directive (EU) 2016/1148 (NIS 2 Directive), 14 December 2022, http://data.europa.eu/eli/dir/2022/2555/oj.

24. PIC/S Guide to Good Manufacturing Practice for Medicinal Products, Annex 15 Qualification and Validation, 25 August 2023, Pharmaceutical Inspection Co-operation Scheme (PIC/S), www.picscheme.org.

25. PIC/S Guide to Good Practices for the Preparation of Medicinal Products in Healthcare Establishments, 1 March 2014, Pharmaceutical Inspection Co-operation Scheme (PIC/S), www.picscheme.org.

26. US Food and Drug Administration (FDA), www.fda.gov.

27. European Medicines Agency (EMA), www.ema.europa.eu/en.

28. International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.

29. Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation), 4 May 2016, https://eur-lex.europa.eu/eli/reg/2016/679/oj.

30. Health Insurance Portability and Accountability Act (HIPAA) Privacy Rule, US Department of Health and Human Services, www.hhs.gov/hipaa/index.html.

31. World Health Organization (WHO), www.who.int.

32. Clinical Data Interchange Standards Consortium (CDISC), www.cdisc.org.

33. Health Level Seven International (HL7), www.hl7.org.

34. ASTM International, West Conshohocken, PA, www.astm.org.

35. ISA/IEC 62443 Series of Standards, International Society of Automation (ISA), www.isa.org/standards-and-publications/isa-standards/isa-iec-62443-series-of-standards.

36. Plattform Industrie 4.0, “Reference Architectural Model Industrie 4.0 (RAMI4.0) – An Introduction,” 8 September 2018, www.plattform-i40.de/IP/Redaktion/EN/Downloads/Publikation/rami40-an-introduction.html.

37. *ISPE Baseline® Pharmaceutical Engineering Guide, Volume 5 – Commissioning and Qualification,* International Society for Pharmaceutical Engineering (ISPE), Second Edition, June 2019, www.ispe.org.

38. ICH Harmonised Guideline: *Pharmaceutical Development – Q8(R2),* Step 4, 1 August 2009, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), www.ich.org.

39. International Organization for Standardization (ISO), www.ISO.org.

40. Allotrope Foundation, www.allotrope.org.

41. MESA International, “B2MML,” Accessed 8 October 2025, https://mesa.org/topics-resources/b2mml/.

42. IEC 62541-5:2020 OPC Unified Architecture – Part 5: Information Model, International Electrotechnical Commission (IEC), www.iec.ch.

43. Eclipse Foundation, https://sparkplug.eclipse.org.

44. SPIRIT-CONSORT Group, www.consort-spirit.org.

45. Heeren, E., Wassink, A. J., Nalluri PhD, V. R., Niderhauser, S., “Methodology to Define a Pharma 4.0™ Roadmap,” *Pharmaceutical Engineering,* May/June 2023, International Society for Pharmaceutical Engineering (ISPE), www.ispe.org.

46. Winter, J., “The Difference Between IT and OT,” 3 June 2024, www.jeffwinterinsights.com/insights/differences-between-it-and-ot.

47. International Society of Automation (ISA), www.isa.org.

48. Cisco Systems, Panduit, and Rockwell Automation, “Deploying a Resilient Converged Plantwide Ethernet Architecture Design and Implementation Guide,” September 2020, https://literature.rockwellautomation.com/idc/groups/literature/documents/td/enet-td010_-en-p.pdf.

49. Gordon, J., “IT/OT Convergence – The Essential Guide,” Industrial Cyber, 9 March 2021, https://industrialcyber.co/analyst-corner/the-essential-guide-to-it-ot-convergence/.

50. BioPhorum, “Digital Plant Maturity Model V3,” 3 November 2023, www.biophorum.com/download/digital-plant-maturity-model-3-0/.

51. *ISPE GAMP® Guide: Artificial Intelligence,* International Society for Pharmaceutical Engineering (ISPE), First Edition, July 2025, www.ispe.org.

52. EudraLex Volume 4 – Guidelines for Good Manufacturing Practices for Medicinal Products for Human and Veterinary Use, Annex 22: Artificial Intelligence (draft guidelines), 7 July 2025, https://health.ec.europa.eu/consultations/stakeholders-consultation-eudralex-volume-4-good-manufacturing-practice-guidelines-chapter-4-annex_en.

53. ISO 23247 Series of Standards: Automation systems and integration – Digital twin framework for manufacturing, International Organization for Standardization (ISO), www.iso.org.

54. Winter, J., “Five Types of Digital Twins,” 14 August 2023, www.jeffwinterinsights.com/insights/five-types-of-digital-twins.

55. EMA Guideline on computerised systems and electronic data in clinical trials, 9 March 2023, European Medicines Agency (EMA) Good Clinical Practice Inspectors Working Group (GCP IWG), EMA/INS/GCP/112288/2023, www.ema.europa.eu/en.

12 Appendix 2 – Glossary

12.1 Acronyms and Abbreviations

ADaM	Analysis Data Model
AGV	Automated Guided Vehicles
AI	Artificial Intelligence
ALCOA++	Attributable, Legible, Contemporaneous, Original, Accurate, Complete, Consistent, Enduring, Available, Traceable
AMR	Autonomous Mobile Robots
API	Application Programming Interface
AR	Augmented Reality
ASRS	Automated Storage and Retrieval Systems
BIM	Building Information Modeling
BPMN	Business Process Management Notation
CAPA	Corrective Actions and Preventive Actions
CDDMS	Clinical Data and Document Management System
CDISC	Clinical Data Interchange Standards Consortium
CFR	Code of Federal Regulations (US)
CMA	Critical Material Attribute
CMMS	Computerized Maintenance Management System
CPP	Critical Process Parameter
CQA	Critical Quality Attribute
CRA	Cyber Resilience Act
CSV	Comma-Separated Value
DCOM	Distributed Component Object Model
DCS	Distributed Control System
DOS	Disk Operating System
DQ	Design Qualification
DTO	Digital Twin of the Organization
EAM	Enterprise Architecture Management
eBR	electronic Batch Record
EDM	Electronic Document Management
ELN	Electronic Laboratory Notebook
EMA	European Medicines Agency
ERP	Enterprise Resource Planning
FAIR	Findable, Accessible, Interoperable, and Reusable

FDA	Food and Drug Administration (US)
FHIR	Fast Healthcare Interoperability Resources
GC	Gas Chromatography
GDPR	General Data Protection Regulation (EU)
GMP	Good Manufacturing Practice
GxP	Good "x" Practice
HIPAA	Health Insurance Portability and Accountability Act (US)
HMI	Human Machine Interface
HPLC	High-Performance Liquid Chromatography
HTTP	Hypertext Transfer Protocol
IaaS	Infrastructure-as-a-Service
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
IEC	International Electrotechnical Commission
IIoT	Industrial Internet of Things
IoT	Internet of Things
IP	Intellectual Property
ISA	International Society of Automation
ISO	International Organization for Standardization
IT	Information Technology
JSON	JavaScript Object Notation
KM	Knowledge Management
KPI	Key Performance Indicator
LIMS	Laboratory Information Management System
LLM	Large Language Model
LMS	Learning Management System
MDM	Master Data Management
MES	Manufacturing Execution System
ML	Machine Learning
MoMa	Mobile Manipulator
MQTT	Message Queuing Telemetry Transport
NCA	National Competent Authorities
OGD	Open Government Data
OEM	Original Equipment Manufacturer
OPC UA	Open Platform Communications Unified Architecture
OT	Operational Technology
PaaS	Platform-as-a-Service

PAT	Process Analytical Technology
PHI	Protected Health Information
PIC/S	Pharmaceutical Inspection Co-operation Scheme
PII	Personally Identifiable Information
PLC	Programmable Logic Controller
PoC	Point of Contact
QA	Quality Assurance
QC	Quality Control
QMS	Quality Management System
QRM	Quality Risk Management
R&D	Research and Development
RAMI	Reference Architectural Model for Industry
ROI	Return on Investment
RTLS	Real-time Location System
SaaS	Software-as-a-Service
SCADA	Supervisory Control and Data Acquisition
SDLC	Software Development Lifecycle
SDTM	Study Data Tabulation Model
SME	Subject Matter Expert
SOP	Standard Operating Procedure
SPC	Statistical Process Control
SPIRIT	Standard Protocol Items: Recommendations for Interventional Trials
SSBI	Self-Service Business Intelligence
SDTM	Study Data Tabulation Model
UNS	Unified Namespace
URS	User Requirement Specification
VR	Virtual Reality
WHO	World Health Organization
WIP	Work-in-Progress
WMS	Warehouse Management System
XML	Extensible Markup Language

701 N. Westshore Blvd., Suite 207, Tampa, Florida 33609 USA
Tel: +1-813-960-2105, Fax: +1-813-264-2816

www.ISPE.org